
Mechanistic Views: A Five-Axis Ontology for Mechanistic Interpretability

Elliot Tower
elliott@elliotttower.ai

Abstract

Mechanistic interpretability has produced circuits, features, causal subspaces, and activation-steered algorithms—each resting on an implicit answer to a prior question: what kind of thing is a mechanism? These are not terminological variants. They differ in ontology, and ontological differences propagate: what a mechanism *is* determines when two descriptions pick out the *same* one, what *evidence* can warrant a claim, and what *inferences* that claim licenses. This paper introduces *mechanistic views*: the explicit background assumptions behind any mechanistic claim. Each view answers five questions: what kind of entity counts as a mechanism (ontology), when two descriptions refer to the same one (identity), what measurements can warrant the claim (evidence), what mathematical language expresses it (formalism), and what phenomenon is being explained (target). The five axes form a dependency chain: ontology constrains identity, which constrains formalism. We argue that many apparent disagreements in the field arise from unstated mismatches between these axes.

A motivating walkthrough shows how a single claim—“head 9.9 is an S-inhibition head”—splits into seven distinct claims under different views, grounded in recent empirical data on circuit overlap, head stability, and prompt specificity. We present an atlas of nine views, classify common methods by their implicit view commitments, and apply the framework to three worked examples. Surveying 17 open problems from seven sources, we find that 8 dissolve once the view is specified, 2 become answerable once the required evidence type is identified, and 7 arise structurally from the limits of the view currently used to study them. The last category is the most important: these are not puzzles waiting for better methods but ceilings that the field’s default operating point—object ontology with activation-patching evidence—cannot transcend.

1 Introduction

Mechanistic interpretability aims to explain what trained neural networks are computing. The field has made this aim concrete: transformer circuits provide a component-level vocabulary for attention heads and MLPs (Elhage et al., 2021); induction heads account for a mechanism involved in in-context learning (Olsson et al., 2022); the indirect object identification (IOI) circuit names the attention heads involved in a specific syntactic task (Wang et al., 2023); sparse autoencoders recover interpretable directions from activation spaces (Bricken et al., 2023; Templeton et al., 2024); causal abstraction methods test whether high-level variables are realized in distributed representations (Geiger et al., 2021; 2024); and weight-space analyses recover inter-layer communication channels and circuit structure from weight matrices alone (Merullo et al., 2024; Elhage et al., 2021).

Yet the word *mechanism* does not mean the same thing in all of these cases. In one paper a mechanism is a set of attention heads; in another it is a functional role such as “name mover”; in another it is a causal subspace recovered by distributed alignment search; in another it is a training-time transition. These are not merely terminological differences. They carry different answers to five basic questions:

- (1) **Ontology**: what kind of entity counts as a mechanism?

-
- (2) **Identity:** when do two descriptions refer to the same mechanism?
 - (3) **Evidence:** what measurements can warrant a claim about a mechanism?
 - (4) **Formalism:** what mathematical language expresses the claim?
 - (5) **Target:** what phenomenon is the mechanism supposed to explain?

These questions are not new. The philosophy of science has studied mechanisms for decades: the Machamer-Darden-Craver (MDC) account defines mechanisms as entities and activities organized to produce a phenomenon (Machamer et al., 2000); Glennan argues that mechanisms ground causal claims (Glennan, 1996); Craver develops mutual manipulability as the criterion for constitutive relevance (Craver, 2007); Bechtel and Richardson study decomposition and localization as discovery strategies (Bechtel & Richardson, 1993); and Illari and Williamson survey the diversity of mechanism concepts across the sciences (Illari & Williamson, 2012). What interpretability adds is precision about the objects: neural network mechanisms are computationally specified, fully inspectable, and amenable to exact intervention. What interpretability lacks is the conceptual infrastructure that the philosophy of mechanisms provides. Williams et al. (2025) independently argue that MI needs philosophical foundations for clarifying concepts, refining methods, and navigating epistemic complexity. This paper draws on that tradition to organize the implicit commitments behind interpretability claims.

Most published work answers these questions implicitly. The answers are recoverable from the methods used and the conclusions drawn, but they are not stated—and because they are not stated, disagreements that are fundamentally ontological are argued as disagreements about experimental design.

Three motivating cases. The object-to-role slide. Wang et al. (2023) establish that head 9.9 is causally necessary for indirect object identification via activation patching. The paper then labels it “an S-inhibition head”—a functional role. These are different claims under different views. The ablation supports the object claim: *this component matters*. The role label requires independent operationalization: does S-inhibition predict behavior on novel syntactic constructions? No such test is reported. This pattern—establishing object identity by patching, then sliding into a role label without independent evidence—is the most prevalent methodological error in interpretability. We call it the **object-to-role slide**, and it recurs throughout the atlas (§4), the diagnostic checklist (Appendix A), and the worked examples (§5).

Cross-model comparison. Merullo et al. (2024) report that the IOI circuit and the Colored Objects circuit in GPT-2 Medium share 78% of their attention heads. Under the object view, this is a striking coincidence that invites mechanistic comparison. Under the role view, it is either confirmation that the same functional roles appear across tasks, or evidence that what was identified as “the IOI circuit” was actually a task-general computational substrate misidentified as task-specific. The two readings have opposite implications for how transferable the IOI findings are—and the choice between them is a view-level commitment, not an empirical question that more patching can resolve.

Safety-relevant inference. Arditi et al. (2024) show that adding a “refusal direction” to activations suppresses refusal behavior. The inference from “this steering vector suppresses refusal” to “this is the refusal mechanism” is a slide from instrumental evidence (the vector works as a behavioral lever) to a structural conclusion (the vector identifies the mechanism). Under the instrumental view, the finding is complete: the direction predicts and controls behavior. Under the object view, the direction must be shown to correspond to necessary components. Under the subspace view, the direction must be stable across distributions. Safety monitoring that relies on steering-vector evidence for mechanistic conclusions—without specifying which view is operative—has an undiagnosed coherence violation at its foundation.

Contributions. This paper makes five contributions:

1. A motivating walkthrough (§2) showing how one claim—“head 9.9 is an S-inhibition head”—reads as seven distinct claims under seven views, grounded in recent empirical data on circuit overlap (Merullo et al., 2024), head stability (Bali et al., 2026), prompt specificity (Franco et al., 2026), and cross-model congruence (Sun & Toneva, 2026).

-
2. A definition of a *mechanistic view* as five interlocking commitments—ontology, identity, evidence, formalism, and target—with coherence conditions linking them (§3).
 3. An atlas of nine views for mechanistic interpretability, with full axis assignments, evidence standards, discriminating experiments, and failure modes (§4).
 4. The determination chain—ontology implies identity implies formalism—with a five-axis methods mapping showing how common tools carry implicit view commitments (§4.10, §4.12).
 5. A survey of 17 open problems from seven sources, decomposed by view: 8 dissolve once the view is specified, 2 become answerable once the required evidence type is identified, and 7 arise structurally from the limits of the view currently used to study them (Appendix D).

These are applied through three worked examples (§5), six recurring decision points (§6), and cross-view promotion conditions (§8).

Relationship to Marr’s levels. The most well-known framework for levels of analysis in computational science is Marr’s tri-level account (Marr, 1982): a computational level (what is computed and why), an algorithmic level (what representation and process), and an implementational level (how the algorithm is physically realized). Marr’s levels organize *explanatory targets*—they ask what a system does, how it does it, and what substrate realizes the process. The five axes here organize *ontological commitments*—they ask what kind of object the mechanism is, when two are the same, and what evidence can support the claim. Marr’s levels do not distinguish views: two researchers at the same Marr level can hold different mechanistic views (object vs. subspace vs. structural). The nine views are complementary to Marr’s hierarchy, not a refinement of it: both are needed, and neither subsumes the other.

2 From Claim to View: One Head, Seven Readings

Before defining views formally, we demonstrate why they matter by taking a single, well-known mechanistic claim and reading it under each view. The claim is:

Head 9.9 in GPT-2 Small is an S-inhibition head in the indirect object identification circuit.

This sentence appears unremarkable. It names a component (head 9.9), assigns it a role (S-inhibition), and places it in a context (the IOI circuit). Yet it carries at least six distinct commitments, and recent empirical work shows that they come apart.

Object view. Under the object view, the claim is about a specific computational unit: attention head 9.9 in a specific model. The evidence is activation patching: ablating head 9.9 degrades IOI performance. The claim is precise within GPT-2 Small and trivially fails cross-model—head 9.9 in Pythia-160M is a different object. Cross-task component reuse (Merullo et al., 2024) shows that head 9.9 appears in circuits for at least four other tasks, raising the question of whether its role in IOI is its essential function or one of many.

Role view. Under the role view, the claim is about a function: S-inhibition, the suppression of attention to the subject token so the model can attend to the indirect object. The component is incidental—any head performing S-inhibition is the same mechanism. Tigges et al. (2024) found that the same algorithmic roles (S-inhibition, name moving, duplicate-token detection) persist across training checkpoints and model scales, even when the specific heads occupying those roles change. Under the role view, this is expected: the role is the mechanism, and the component is a variable realization. Under the object view, this is a different mechanism at each scale.

Subspace view. Under the subspace view, the relevant entity is not head 9.9 but the subspace of the residual stream at layer 9 that encodes the subject–object distinction. Bali et al. (2026) measured attention-head stability across random seeds and found mid-layer head assignments unstable (stability ≈ 0.70), while

residual-stream subspaces remain stable as measured by centered kernel alignment. Under the subspace view, this result makes the object-level description unreliable and the subspace-level description robust: the computation is stable, but which head implements it is not.

Structural view. Under the structural view, the relevant object is the gauge-invariant information flow: the composition score $\|W_{O,9.9}W_{K,v}\|_F$ for downstream heads v that read from head 9.9. A computation-preserving reparameterization (e.g., rotating W_Q and W_K by R and R^{-1}) changes the individual weight matrices but not the composition score. If the activation-patching result survives such reparameterizations, the claim is structural; if it does not, it is basis-dependent.

Process view. Under the process view, the question is *when* head 9.9 acquired the S-inhibition function. Olsson et al. (2022) showed that induction heads form via phase transitions at specific training steps, with composition scores between heads increasing sharply at the transition. Does S-inhibition form similarly? The answer requires checkpoint analysis that neither the original IOI paper nor subsequent replications provide. Under the process view, the mechanism is the formation trajectory, not the final-checkpoint circuit.

Contrastive view. Under the contrastive view, the S-inhibition claim is defined relative to a foil: head 9.9 suppresses attention to the subject *compared to what*? Franco et al. (2026) showed that even within IOI, different prompt templates (ABBA vs. BABA) activate different circuit structures. The signals flowing through head 9.9 differ between templates—negative cosine similarity in one, near-zero in another. The contrastive view says these are different mechanisms, because the foil (which template) changes what the head is contrasted with.

Perspectival view. Under the perspectival view, each of the above is a partial projection. Activation patching reveals the object-level shadow; DAS reveals the subspace-level shadow; weight-space composition scores reveal the structural shadow. Sun & Toneva (2026) found that within model families, interpretive equivalence is high (congruence 0.89–0.99), but across families it collapses to 0.13. The perspectival view interprets this as method-relative structure: a single method identifies a mechanism within a family, but the identification does not transport to a different family because the measurement context has changed.

What this walkthrough shows. The sentence “head 9.9 is an S-inhibition head” is not one claim. It is at least seven, depending on whether the speaker means a specific component (object), a functional role (role), a causal subspace (subspace), a gauge-invariant structure (structural), a formation event (process), a foil-relative difference (contrastive), or a method-projected observation (perspectival). The empirical data—Merullo’s 78% overlap, Tigges’s cross-scale persistence, Bali’s head instability, Franco’s prompt specificity, Sun’s within-family congruence—each supports a different reading and undermines others.

The framework developed in the following sections provides the vocabulary to state which reading is intended, check whether the evidence supports it, and identify when a disagreement between researchers is actually a mismatch between unstated views.

3 Mechanistic Views

3.1 What a Mechanistic View Is

A mechanistic view is a background commitment that a researcher makes, explicitly or implicitly, when they ask what a mechanism is. When made explicit, a view answers five questions:

- (i) **Ontology:** what kind of entity counts as a mechanism? (Concrete components, functional roles, causal subspaces, gauge-invariant structures, formation processes, etc.)
- (ii) **Identity:** when do two descriptions refer to the same mechanism? (Component overlap, role equivalence, subspace proximity, gauge-orbit membership, etc.)

-
- (iii) **Evidence:** what measurements can support what claims? (Activation patching, DAS/IIA, weight-space scores, checkpoint analysis, etc.—and critically, which claims each measurement type can and cannot support.)
 - (iv) **Formalism:** what mathematical language expresses the claim? (Directed graphs, Grassmannian geometry, fiber bundles, dynamical systems, etc.)
 - (v) **Target:** what phenomenon is the mechanism supposed to explain? (A specific behavior, a functional class, a representational variable, a computation class, etc.)

The identity criterion should be an equivalence relation (reflexive, symmetric, transitive). This is not automatic: some informal notions of mechanism identity in the literature are not transitive. For example, saying mechanisms A and B are “related” by shared components, and B and C are “related” by shared function, does not imply A and C are the same mechanism. Requiring transitivity forces precision.

3.2 Coherence

Not all combinations of answers are coherent. A view is coherent when its five components fit together:

- (i) **Identity is grounded:** the identity criterion is definable in terms of the ontology and the formalism. (A view whose identity criterion involves objects not in its own ontology is incoherent.)
- (ii) **Evidence tracks identity:** measurements should not distinguish objects that the identity criterion declares equivalent. (Evidence that discriminates within an identity class is tracking a finer-grained ontology than the one declared.)
- (iii) **Formalism is expressive:** every ontological object has a representation in the formalism, and identity can be checked within it.

When a view is incoherent—when its evidence distinguishes things its identity criterion says are the same—the result is unresolvable disagreement. No additional experiment of the same type can help: each new measurement either agrees with the identity criterion (offering no information about the discrepancy) or disagrees (adding another conflict).

This is not an abstract concern. A common pattern in interpretability is to declare that mechanisms are gauge-invariant structures (structural view ontology) but then to use activation-patching results as evidence of mechanism presence without checking gauge invariance. If the patching result changes under a reparameterization that preserves computation, it is distinguishing objects that the structural view says are identical—and no amount of additional activation-patching evidence can resolve this mismatch. The disagreement between patching advocates and DAS advocates often has this structure: the participants hold different implicit views, so they are literally arguing about different kinds of objects.

3.3 The Five Axes

We organize the five components into named axes. The axes are not independent: as coherence requires, choices on one axis constrain the others.

Ontology (O). What kind of entity counts as a mechanism (Quine, 1948)? Options include: concrete model components (heads, neurons, MLP sublayers, SAE dictionary elements), functional roles (descriptions of what a component does independently of which component does it), causal subspaces (linear or nonlinear submanifolds of the residual stream that mediate a causal variable), gauge-invariant structures (equivalence classes of weight configurations under symmetries that preserve the computation), temporally extended processes (formation pathways or execution trajectories), points in a stratified mechanism space, measurement perspectives, and predictive models.

Identity ($\sim$). When do two mechanism descriptions denote the same mechanism? Options include: component overlap ($x \sim y$ iff they name the same set of heads or neurons), role equivalence ($x \sim y$ iff they specify the same functional input–output mapping), subspace proximity ($x \sim y$ iff the principal angles between the causal subspaces are all below a threshold θ , i.e. geodesic distance on $\text{Gr}(k, d)$ is small), gauge-orbit membership ($x \sim y$ iff there exists a computation-preserving symmetry g with $g \cdot x = y$), and dynamical-basin membership ($x \sim y$ iff the formation trajectories converge to the same attractor).

Evidence (E). What measurements can support what claims? Key distinctions: activation patching establishes causal relevance of components under a fixed distribution; DAS/IIA with linear alignment constraints measures causal subspace identity; weight composition scores measure structural information flow independent of any input distribution; checkpoint analysis establishes formation dynamics; multi-method robustness tests perspectival convergence. The formal epistemology of evidence accumulation (Earman, 1992; Howson & Urbach, 1989) suggests that independent evidence converges faster than correlated evidence—a principle that grounds the triangulation requirement in the perspectival view.

Formalism (F). What representational language is appropriate? Directed graphs (circuits); functional decompositions (roles); Grassmannian geometry $\text{Gr}(k, d)$ (subspaces); fiber bundles and gauge groups (structural view); dynamical systems and phase portraits (process view); Whitney stratifications (stratified view); measurement algebras (perspectival view).

Target (T). What is the mechanism supposed to explain? A specific behavioral output on a specific distribution; a functional class across models; a representational variable across inputs; a computation class across parameterizations; a formation event; a resolution-relative phenomenon; or a behavioral prediction.

4 An Atlas of Views

We identify nine views. Six are *descriptive*: they characterize views already operative in published work (Object, Role, Subspace, Process, Instrumental, Contrastive). Three are *programmatic*: they articulate views that are implicit in emerging research directions but do not yet have widely adopted methods (Structural, Stratified, Perspectival). Table 1 gives the full axis assignments. The views are not mutually exclusive: a single paper may use several, and convergence across views constitutes stronger evidence than any single view. The point is to distinguish them so that each view’s commitments and limitations can be assessed independently.

View	Ontology (O)	Identity ($\sim$)	Evidence (E)	Formalism (F)	Target (T)
Object	Concrete part: head, neuron, feature	Component overlap	Ablation, activation patching	Directed graph	Specific behavior
Role	Functional role in computation	Role equivalence (same I/O mapping)	Role-specific causal tests, cross-model	Functional decomposition	Functional class
Subspace	Causal subspace of residual stream	Same projector; $d_{Gr} < \theta$	DAS/IIA (linear), subspace stability	$Gr(k, d)$	Representational variable
Structural	Gauge-invariant relational structure	Gauge-orbit membership	Holonomy, comp. scores, reparameterization	Fiber bundle quotient	Computation class
Process	Formation trajectory or execution regime	Same trajectory type or dynamical basin	Training checkpoints, formation knockouts	Dynamical system	Mechanism origin
Stratified	Point in stratified mechanism space	Same stratum + local equivalence	Dimensionality tests, localization diagnostics	Whitney stratification	Resolution-relative phenomenon
Perspectival	Projection of mechanism onto context	Cross-method coherence	Multi-method robustness	Measurement algebra	Method-relative observation
Instrumental	Useful predictive or interventional model	Predictive equivalence	Forecast accuracy, intervention utility	Model theory	Behavioral prediction
Contrastive	Difference relative to a foil	Same contrastive pattern across foil set	Foil-varied patching, contrastive ablation	Contrastive explanation	Why P rather than Q

Table 1: The nine mechanistic views with full axis assignments.

4.1 Object View

The object view treats mechanisms as concrete model components: specific attention heads, neurons, MLP sublayers, or sparse autoencoder dictionary elements. It is the implicit view in circuit diagrams. The IOI circuit as reported by Wang et al. (2023)—26 attention heads organized into functional groups—is an object-level description: the claim is about those specific computational units.

Identity. Two circuit descriptions refer to the same mechanism iff they name the same components. This is precise within a model but trivially fails cross-model: a head in GPT-2 Small is not an object in Pythia-160M.

Evidence. Ablation and activation patching are the natural evidence types. Removing a component’s activation and observing behavioral degradation establishes causal necessity; replacing it with a counterfactual activation establishes causal sufficiency relative to that counterfactual. The method is strongest when the component set is stable across prompt distributions and the ablation effect is large and specific.

Failure modes. *Role inflation:* a component is identified by patching, then labeled with a role (“name mover,” “induction head”) that implies richer mechanistic content than the object-level evidence supports. The component identity is established; the role is a hypothesis. *Distribution sensitivity:* the circuit identified by patching on one distribution may differ substantially on another, meaning the claimed object-level mechanism is distribution-relative.

Discriminating experiment. To test whether the object view is appropriate: ablate the identified component set and test whether behavioral degradation is (a) specific to the behavior the mechanism is supposed to implement and (b) stable across prompt distributions. If (b) fails, the object-level description is distribution-relative and a role or subspace description may be more appropriate.

Compositional variant. A natural refinement of the object view defines mechanisms not by component indices but by the computation graph they implement: each edge is a tensor operation, and two circuits are the same if their computation graphs are isomorphic. This is strictly more abstract than component overlap (it ignores which head occupies each node) but more concrete than the role view (it preserves graph structure, not just input–output behavior). Transcoder-based circuit analysis implicitly adopts this variant. We treat it as a variant of the object view rather than a separate view because it shares the same evidence types and failure modes; the difference is in the identity criterion only.

4.2 Role View

The role view treats mechanisms as functional roles that can be realized by different components—the role is a universal that can be multiply instantiated (Lewis, 1972). “Name mover” is a role: it describes what a component does (copy a name token to the output position), not which head does it. The role view explains why functionally equivalent components across different models can be meaningfully compared.

Identity. Same functional input–output role, possibly in different components or models. Role identity requires a specification of the role that is independent of the discovery procedure (i.e., not merely “does what this head does”) and tests that go beyond the ablation that revealed the component.

Evidence. Role-specific predictions that are not entailed by mere causal relevance. For a name-mover role claim: the weight-space signature of W_{OV} should encode a copying operation; the head should perform this function on novel prompts; a head in a different model with the same W_{OV} signature should earn the same label. The object-to-role slide is a common source of overclaiming: establishing that head 9.9 is causally necessary for IOI does not establish that it is a name mover—the ablation evidence supports the object claim (this component matters) but not the role claim (this component performs a specific function) until the role is independently operationalized and tested.

Failure modes. *Role proliferation:* a new role label is coined for each component discovered, re-describing the data without adding explanatory content. *Underdefined roles:* role labels are informal enough to be true of any component, making them unfalsifiable.

Discriminating experiment. Take the role specification and apply it to a model in which the original component has been ablated. If the role is real, either another component should take over (backup mechanism) or the behavior should degrade in a specific role-consistent way. If no such prediction is confirmed, the role label has not been independently validated.

4.3 Subspace View

The subspace view treats mechanisms as low-dimensional causal subspaces of the residual stream. The key observation is that distributed alignment search (Geiger et al., 2021; 2024) identifies not a component but a rotation of the activation space: the causal variable is encoded in a subspace that may span many components. If alignment matrix Q and its rotation QR both achieve the same interchange intervention accuracy (IIA), the relevant object is the subspace $\text{span}(Q)$, not the matrix Q .

Identity. Two subspace mechanisms are the same iff they project onto the same subspace of $\mathbb{R}^d$: same projector $QQ^\top$, or equivalently small geodesic distance on $\text{Gr}(k, d)$. This is basis-invariant and can be meaningful cross-model when corresponding residual streams can be aligned.

Evidence. The subspace view requires evidence that a specific subspace mediates a causal variable, not merely that a search over subspaces finds one that works. Native subspace evidence includes: SVD or eigen-decomposition of weight matrices recovering the subspace without activation data; Grassmannian distance between subspaces recovered by independent methods (weight vs. activation); and subspace stability across prompt distributions. DAS and IIA are often cited as subspace evidence, but as Table 2 shows, DAS is more precisely a Role-view method with a borrowed subspace parameterization: it validates by intervention

success (role equivalence), not by Grassmannian distance. DAS can *discover* a candidate subspace, but confirming that it is the causally relevant subspace—rather than merely a subspace that permits successful interchange—requires convergent evidence from weight-space analysis.

Alignment class constraints. Sutter et al. (2025) showed that unrestricted nonlinear alignment maps achieve 100% IIA even on randomly initialized models, making DAS vacuous without structural constraints on the alignment class. Subspace-view claims therefore require either linear alignment or alignment constrained to respect the transport structure induced by the weight matrices. Subject to this constraint, subspace stability (eigenvalue gap between the causal subspace and its complement) is the natural diagnostic. This constraint recurs throughout the atlas: it limits the evidential range of DAS (Section 4.12), appears in the diagnostic checklist (Appendix A), and is the primary threat to subspace claims in the worked examples.

Failure modes. *SAE confusion:* sparse autoencoder features are dictionary elements that minimize reconstruction loss; causal subspaces are directions that mediate specific causal paths. These are different objectives and can disagree. An SAE feature is a hypothesis about a one-dimensional subspace mechanism, not a confirmed subspace claim. *Unconstrained alignment vacuousness* (see above): reporting high IIA without specifying the alignment class allows any model to be assigned any causal structure.

Discriminating experiment. Verify that the identified subspace is stable: the subspace recovered by DAS on a held-out distribution should have small geodesic distance from the training-distribution subspace. Also confirm that an aligned basis-change (within the subspace) does not change the IIA, while a rotation that takes vectors out of the subspace does reduce it.

4.4 Structural View

The structural view treats mechanisms as equivalence classes of implementations under transformations that preserve the computation. The relevant object is not a head index, a basis vector, or a specific projection—it is the orbit of implementations related by the symmetry group of the transformer’s weight space. Two weight configurations that implement the same computation via different parameterizations are the same mechanism under this view.

The natural formalism borrows from differential geometry: gauge orbits, transport maps, and holonomy groups (Nakahara, 2003). This view is largely programmatic—the formalism exists, but systematic empirical application to neural networks does not yet.

Identity. Gauge-orbit membership: $x \sim y$ iff there exists a computation-preserving symmetry g (such as a scale transformation, a rotation of a head’s query-key subspace paired with an inverse rotation of its output, or a permutation of neurons) such that $g \cdot x = y$. Holonomy fingerprints provide a finer criterion: the holonomy group of the transport map induced by a mechanism’s weight matrices is an invariant that does not change under gauge transformations.

Evidence. Gauge-invariant measurements: weight-space composition scores $\|W_{O,u}W_{K,v}\|_F$ measure information flow capacity independently of basis choice (Elhage et al., 2021; Merullo et al., 2024); holonomy measurements; and reparameterization-stability tests (check that a result survives computation-preserving rescalings of the weights).

Failure modes. *False gauge invariance:* claiming structural equivalence without verifying that the symmetry actually preserves the computation. *Architectural prior confound:* structural features that discriminate trained from randomly initialized models may be detecting architecturally privileged positions rather than learned structure.

Discriminating experiment. Apply a known computation-preserving symmetry (e.g., scale by λ , counter-scale by λ^{-1}) to the weights of a component and retest the activation-space evidence. If the

activation-space result changes, the original result was not gauge-invariant and is not a structural-view claim.

4.5 Process View

The process view treats mechanisms as temporally extended. A mechanism is not merely a final-checkpoint artifact but a formation process, an execution trajectory, or a dynamical regime—what Darden calls “reasoning about mechanisms in the making” (Darden, 2006). This view is natural for phenomena where the how of learning is itself mechanistically relevant: the induction head mechanism, for instance, forms via a phase transition at a specific training step (Olsson et al., 2022)—this is process-level structure that is not recoverable from the final checkpoint alone.

Identity. Same trajectory type or dynamical basin. Two processes are the same mechanism if they follow the same formation pathway, occupy the same attractor, or exhibit the same phase-transition structure. This notion of identity is less developed than component or subspace identity, and formalizing it is an open problem.

Evidence. Checkpoints, gradient flow, loss-curve structure, and formation knockouts. A mechanism claim under the process view cannot be supported by final-checkpoint analysis alone. The induction head formation result provides a clear example: the phase change at a specific training step is visible in the composition score evolution, and this is process-level evidence distinct from any claim about the final circuit (Olsson et al., 2022).

Failure modes. *Process-circuit conflation:* the final-checkpoint circuit and the formation process are distinct objects; the circuit does not determine the process. Two models may arrive at identical final circuits via very different training trajectories. *Training-detail overfitting:* a process described precisely enough to be informative may be specific to a particular optimizer, learning rate schedule, or dataset.

Discriminating experiment. Run the same architecture with different training hyperparameters (learning rate, batch size, optimizer) and compare the formation dynamics. If the final circuit is similar but the formation trajectory differs, the process-level claim is hyperparameter-sensitive and should be stated at the appropriate level of generality.

4.6 Stratified View

The stratified view treats mechanism space as organized into strata: point-localized objects, one-dimensional features, higher-dimensional subspaces, nonlinear manifolds, and fully distributed structures. Rather than forcing a binary choice between localized and distributed, the stratified view asks which stratum the evidence supports and whether the appropriate description changes with measurement resolution.

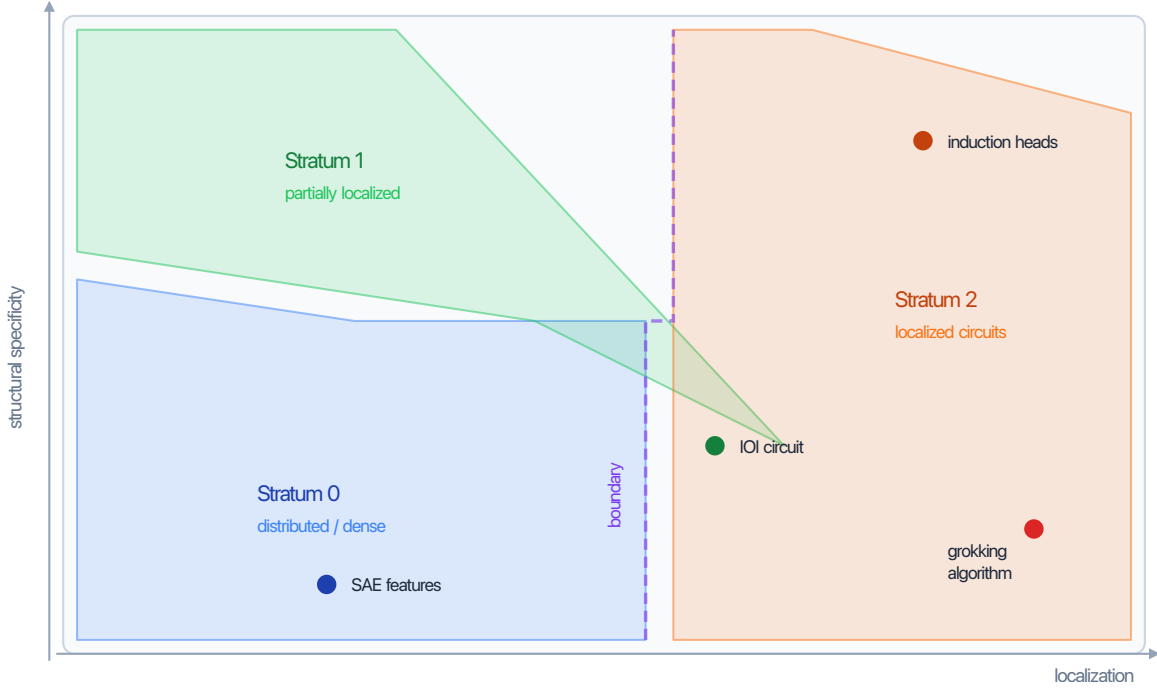


Figure 1: Schematic of mechanism space as a stratified manifold. Each stratum is a smoothly varying subset distinguished by the localization and structural specificity of the mechanism. Boundaries between strata are lower-dimensional singular sets. Empirical examples are placed roughly by the evidence reviewed in Section 5; exact positions are schematic, not quantitative.

Figure 1 illustrates the key distinctions schematically. The natural formalism is Whitney stratification (Whitney, 1965; Golubitsky & Guillemin, 1973): a decomposition of mechanism space into smooth manifold strata that are partially ordered by containment of their closures. The stratified view has a direct analogy in physics: the renormalization group (RG) describes how the effective description of a system changes with measurement resolution (Wilson, 1971; Goldenfeld, 1992). RG fixed points correspond to stable strata; relevant operators correspond to structure that survives coarse-graining; universality classes correspond to mechanisms that appear identical at a given resolution despite microscopic differences. The stratified view imports this resolution-dependence into interpretability.

A mechanism’s stratum is determined by its dimensionality, its localizability in specific components, and the topology of its support in the residual stream.

Identity. Same stratum and same local equivalence class within the stratum. Two descriptions are the same mechanism if they name the same stratum and agree on local structure within that stratum.

Evidence. Dimensionality tests (effective rank of the causal subspace), localization diagnostics (is the mechanism’s effect concentrated in a small component set or distributed?), and stratum-transition tests (does the apparent mechanism change stratum when the measurement resolution changes?).

Failure modes. *Stratum misidentification from method artifacts.* A genuinely distributed mechanism may appear localized to a single component if the ablation method captures only the dominant direction. Conversely, a localized mechanism may appear distributed if the measurement method has high background noise. The stratified view does not resolve this automatically—it provides the language to state it precisely.

Discriminating experiment. The stratified view predicts that measurement resolution affects the apparent stratum. Test this by comparing the effective dimensionality of the causal subspace recovered by DAS at different hook points and with different alignment constraints. Consistent stratum across resolutions supports a stratum assignment; stratum-change suggests either a higher-stratum mechanism being partially recovered or a measurement artifact.

4.7 Perspectival View

The perspectival view treats methods as revealing partial projections of mechanisms rather than the mechanism in itself. Disagreement between methods is not automatically a failure; it may reveal the method-relative structure of what is being measured. This view is philosophically related to Massimi’s perspectival realism (Massimi, 2022): the claim is not that mechanisms are relative to methods (that would be instrumentalism), but that scientific methods are partial perspectives on a method-independent mechanistic structure. The key distinction—between perspectival *evidence* (measurements are always from a vantage point) and perspectival *truth* (what’s true depends on who measures it)—maps directly onto the primary failure mode: conflating the method-relativity of evidence with the claim that mechanisms are themselves method-relative objects.

Identity is coherence across perspectives: two descriptions refer to the same mechanism if they are mutually consistent and jointly interpretable as projections of the same underlying structure. Evidence is multi-method robustness, formalized as convergence across structural (weight), causal (activation/intervention), and dynamic (training-time) evidence families. The discriminating experiment is cross-method consistency: test whether a patching-supported claim is also supported by weight-space composition scores. If not, a third method adjudicates.

4.8 Instrumental View

The instrumental view treats mechanisms as useful predictive or interventional models, warranted by forecast accuracy and intervention utility regardless of whether they identify a real internal object. Identity is predictive equivalence: two descriptions are the same mechanism if they make the same predictions about all interventions and behaviors. This has the weakest ontological commitments and is therefore the easiest to satisfy—but also the most limited. The primary failure mode is *safety-relevant overreach*: claims that require strong ontological conclusions (“the model is pursuing a goal”) cannot be supported by predictive utility on the training distribution alone. The discriminating experiment: test whether the mechanism label generates novel predictions beyond the behavior used to identify it.

4.9 Contrastive View

The contrastive view treats mechanisms as defined relative to a foil: a mechanism is what a component or subspace does in condition *A* compared to condition *B*. This is how most interpretability experiments actually work in practice—the IOI circuit is studied by contrasting indirect-object completion with subject completion, not by characterizing what heads do in absolute terms—but the contrastive structure is rarely made explicit as an ontological commitment.

The contrastive view draws on a well-developed tradition in philosophy of science. Van Fraassen’s pragmatic theory of explanation holds that explanations are always contrastive: we explain why *P* rather than *Q*, not why *P* simpliciter (van Fraassen, 1980). Lipton’s contrastive epistemology extends this to inference: we infer the cause of *P*-rather-than-*Q*, and different foils can yield different (and jointly correct) causal explanations (Lipton, 1990).

Identity. Two mechanism descriptions refer to the same mechanism iff they produce the same contrastive pattern across the same foil set. A “name mover” mechanism identified by contrasting IO with S tokens is a different contrastive mechanism from one identified by contrasting IO tokens with random tokens, even if both identify the same head—because the foil determines what aspect of the head’s behavior is being explained.

Evidence. Contrastive evidence requires specifying both the target condition and the foil. Activation patching with a specific counterfactual is inherently contrastive (the counterfactual is the foil); mean ablation is less so (the foil is an average, not a specific alternative). The strongest contrastive evidence comes from systematic variation of the foil set: if the mechanism’s effect is stable across foils within a class but changes across foil classes, the contrastive boundary reveals the mechanism’s scope.

Failure modes. *Foil dependence without acknowledgment:* reporting a mechanism claim without specifying the foil, then treating it as an absolute characterization. *Foil cherry-picking:* choosing the foil that produces the cleanest result without testing alternatives.

Discriminating experiment. Take a published mechanism claim and vary the foil systematically. If the circuit identified for IOI changes when the foil is changed from “repeat the subject” to “output a random name” to “output nothing,” the mechanism is foil-relative and should be stated as such.

4.10 The Determination Chain

The five axes are not independent. In practice, the relationship is mostly a chain of determination:

$$\text{Ontology} \longrightarrow \text{Identity} \longrightarrow \text{Formalism}$$

Choosing what kind of thing a mechanism *is* determines when two mechanisms are the *same*, which in turn determines what *mathematical structure* can express the claim. If mechanisms are causal subspaces, identity is geodesic proximity on $\text{Gr}(k, d)$, and the formalism must include Grassmannian geometry. If mechanisms are gauge-invariant structures, identity is gauge-orbit membership, and the formalism must include fiber bundles and holonomy groups. The chain is mostly one-to-one: each of the nine ontological commitments in the atlas implies a specific identity criterion, which implies a specific ambient formalism (Table 1).

Evidence and target are constrained by the ontology–identity pair but not fully determined by it: a subspace claim can be supported by activation-space evidence (DAS), weight-space evidence (SVD), or both, and the target is chosen by the researcher. The determination chain therefore has two free parameters once the ontology is fixed.

This structure explains why the most common coherence violations involve mismatches between the first three axes: using a subspace formalism (Grassmannian distance) while holding an object ontology (component overlap identity), or using an object evidence type (activation patching) to support a structural claim (gauge-orbit identity). The chain makes these mismatches visible.

4.11 View Families and Ontological Commitment

The nine views group into families based on their shared structure (Figure 2):

- **Identity family** (Object, Role): mechanisms are identified by what they are (components) or what they do (functions). Lowest mathematical overhead; most of the current literature operates here.
- **Mathematical family** (Subspace, Structural): mechanisms live in geometric spaces and identity is defined by distances or orbits in those spaces. Requires differential geometry.
- **Process family** (Process, Stratified): mechanisms are temporally extended or resolution-dependent. Requires dynamical systems theory or stratification theory.
- **Methodological singleton** (Perspectival): no single method reveals the mechanism; identity requires cross-method coherence.
- **Pragmatic singleton** (Instrumental): mechanisms are useful models; identity is predictive equivalence.
- **Contrastive singleton** (Contrastive): mechanisms are defined relative to a foil; identity requires matching contrastive patterns.

One natural (though not unique) ordering of the views is by increasing ontological commitment—how much they claim about the existence and nature of the mechanism independent of any measurement procedure:

Instrumental < Contrastive < Perspectival < Object < Role < Subspace < Structural < Process < Stratified

This ordering is approximate: one could argue that Role has less ontological commitment than Object (since a role is multiply realizable while a component identity is tied to a specific model), and the relative positions of Process and Structural depend on whether one weights temporal or algebraic structure as “more committed.” What is clear is the endpoints: the instrumental view requires only predictive utility, while the stratified view requires evidence across multiple strata and measurement resolutions. Most published interpretability work operates at the Object or Role level. Moving to higher-commitment views requires explicitly stating which view is operative and supplying the corresponding evidence.

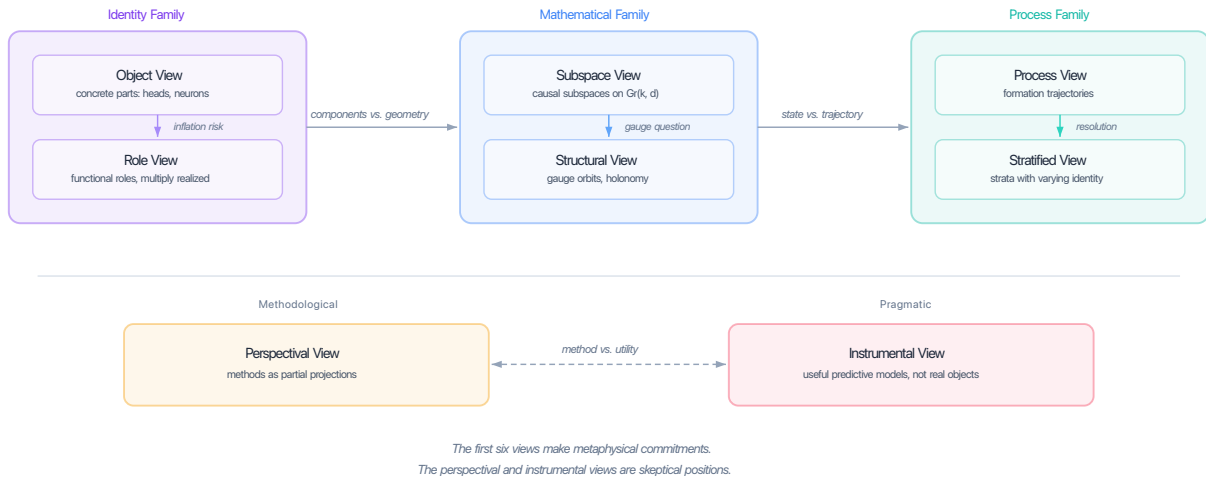


Figure 2: The nine views grouped into three families (identity, mathematical, process) plus three singletons (methodological, pragmatic, contrastive), ordered by increasing ontological commitment. Within each family, the internal arrow marks the principal decision point between the two views.

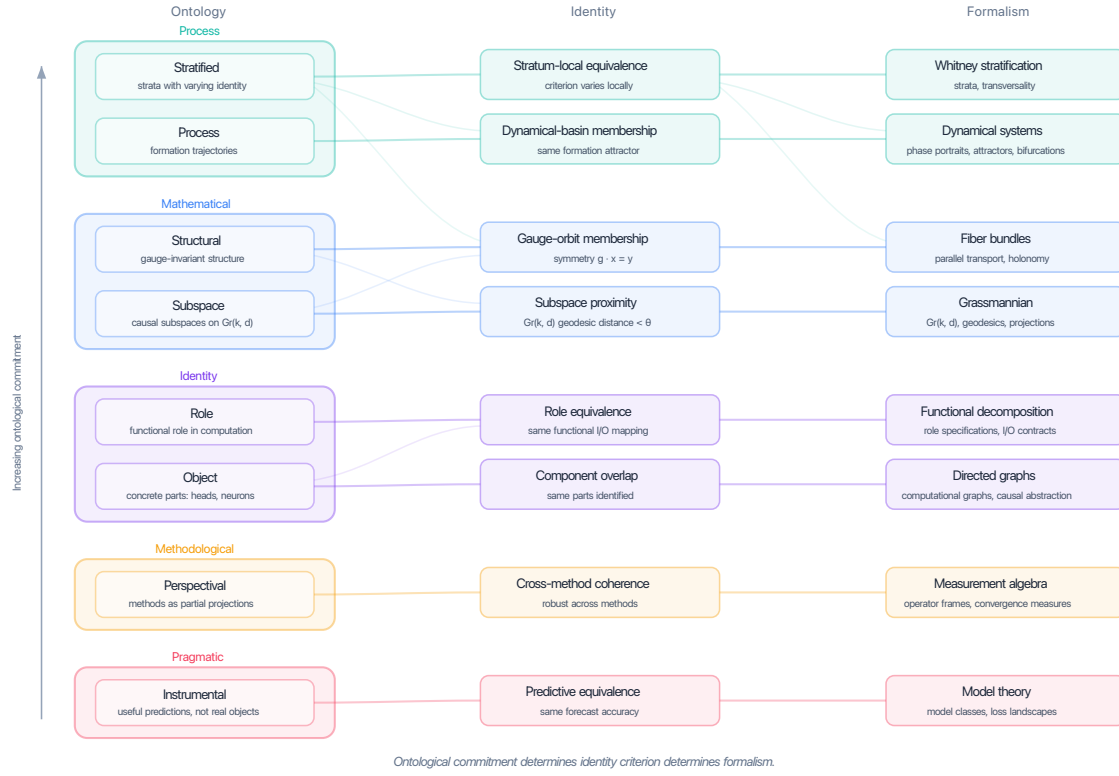


Figure 3: The determination chain: ontology determines identity determines formalism. Each view’s ontological commitment (left) implies a specific identity criterion (center), which implies a natural mathematical formalism (right). Primary connections are shown in bold; secondary cross-connections are shown faintly. Increasing ontological commitment runs from bottom to top.

4.12 Methods Mapping

Each commonly used interpretability method carries implicit commitments across all five axes. Table 2 makes these explicit.

Method	Ontology	Identity	Evidence	Formalism	Target
Activation patching	Object	Component overlap	Activations	Directed graph	Task circuit
Path patching	Object	Component overlap	Activations	Directed graph	Task circuit
ACDC	Object	Component overlap	Activations	Directed graph	Task circuit
EAP	Object	Component overlap	Act. + Wt.	Directed graph	Task circuit
Ablation	Object	Component overlap	Activations	Directed graph	Importance
DAS / IIA	Role [†]	Role equivalence	Activations	Subspace (bordered)	Concept
Causal scrubbing	Role	Role equivalence	Activations	Causal graph	Alignment
Linear probing	Role	Role equivalence	Activations	Linear classifier	Detection
SAE features	Object	Component overlap	Activations	Dictionary	Feature catalog
Logit / tuned lens	Object	Component overlap	Activations	Linear proj.	Layer readout
SVD of weights	Subspace	Subspace prox.	Weights	$\text{Gr}(k, d)$	Decomposition
Composition scores	Structural	Gauge orbit	Weights	Fiber bundle	Info-flow bound
AGOP	Process	Basin membership	Dynamics	Dynamical sys.	Trajectory

Table 2: Five-axis classification of common interpretability methods. Most methods are Object view with activation evidence. [†]DAS uses a subspace *parameterization* (the alignment matrix searches over $\text{Gr}(k, d)$) but Role *identity*: it validates by IIA success, not Grassmannian distance. The ontology is a hybrid—subspace search space, role validation criterion.

Several patterns are visible. First, the field’s default operating point is Object ontology with activation evidence: activation patching, path patching, ACDC, ablation, SAE features, and logit lens all sit here. Second, DAS occupies an interesting intermediate position: it uses a subspace parameterization (the alignment matrix Q searches over subspaces) but validates by interchange intervention accuracy, which is a *role*-level criterion—whether two representations can be swapped without loss of function. The subspace is the search space, not the ontology. Third, no widely-used method operates in the Stratified or Perspectival views—these views generate research programs rather than describing current practice.

5 Worked Examples

Three extended examples show what the atlas does in practice: not just describing views in the abstract, but showing how the same body of evidence is read differently under different views and what each reading implies for follow-up work. IOI illustrates a coherence violation (the object-to-role slide); induction heads illustrate what convergence across views looks like (the existence proof that the framework’s triangulation requirement is satisfiable); and superposition illustrates the framework’s strongest capacity—dissolving an active debate by showing it is a view disagreement rather than an empirical one. Additional case studies (SAE features, gender bias, factual recall, hallucination, grokking, and others) appear in Appendix B.

5.1 Indirect Object Identification

Wang et al. (2023) identify a circuit for indirect object identification (IOI) in GPT-2 Small: 26 attention heads organized into functional classes including S-inhibition heads, name movers, and duplicate-token detection heads.

Object view. The mechanism is those 26 heads. The evidence (activation and path patching, mean ablation) establishes causal necessity and sufficiency relative to the test distribution. Object-level identity is precise within GPT-2 Small and fails across architectures.

The object-level result is solid. What it does not establish: (a) that these heads perform a functional role that would be recognizable in another model, (b) that the relevant causal variables are localized in specific subspaces, (c) that the result is invariant under parameter reparameterizations, or (d) how the circuit formed. Claims (a)–(d) require distinct evidence.

Role view. The four role classes (S-inhibition, name mover, duplicate-token detection, and backup name mover) constitute a role-level description. This is a hypothesis about what the heads do, not a confirmed role claim. To confirm it, each role should be operationalized and tested independently: name movers should copy tokens in novel syntactic constructions; S-inhibition heads should suppress the subject in novel contexts where the subject–object relationship is structurally varied. Until these tests are run, the role labels describe the experimental template more than the mechanism.

Subspace view. The subspace question is where the IO and S tokens, and the inhibitory relationship between them, are represented in the residual stream. DAS with linear alignment constraints would identify the causal subspaces for each variable and test whether they can be independently intervened on. The 26-head object-level description does not resolve this: two different subspace configurations could produce the same patching results.

Structural view. The structural question is which aspects of the circuit are gauge-invariant. Path-patching results that survive computation-preserving reparameterizations of the relevant head weights establish structural facts; those that do not are basis-dependent observations. The holonomy of the circuit’s transport map would provide a structural fingerprint that is invariant across model instances.

Contrastive view. The IOI circuit is inherently contrastive: it characterizes what heads do when distinguishing the indirect object from the subject. A different foil—IO vs. random token, or IO vs. nothing—would identify a different contrastive mechanism, possibly with different component involvement. Franco et al. (2026) confirm this: ABBA and BABA prompt templates, which instantiate different foil structures, activate different circuit topologies within the same model.

Verdicts. *Object view:* the object-level claim is established; the 26-head circuit is causally necessary on the test distribution. *Role view:* role identity is asserted, not confirmed; no independent role operationalization exists. *Subspace view:* not attempted; the causal subspaces have not been identified. *Structural view:* not attempted; gauge invariance has not been tested. *Contrastive view:* the IOI circuit as stated is foil-relative; varying the foil changes the circuit.

5.2 Induction Heads

The induction head result (Olsson et al., 2022) is unusually well-supported, precisely because multiple views converge on the same object.

Object view. Induction heads are identified by activation patching. Their distinctive attention pattern—attending to the token immediately after the previous occurrence of the current token—is established across a large family of transformers. Object-level identity is extended cross-model by demonstrating the pattern holds across model sizes and architectures.

Role view. The induction pattern defines a functional role that is operationalized (it can be tested on any sequence with a repeated token) and confirmed across model families. This is one of the few cases in the literature where role identity is genuinely established rather than hypothesized. The role specification is independent of the component: any head that attends to t_j from position i when token $t_i = t_{j-1}$ is an induction head, regardless of layer or position.

Process view. The formation dynamics are studied across training checkpoints. A phase transition is identified at a specific step in the loss curve, and the composition score between the head and its prefix-matching partner increases sharply at that step (Olsson et al., 2022). This is process-level evidence—it concerns how the mechanism formed, not what it is at the final checkpoint. The formation process is argued to be causally responsible for the in-context learning capacity of the model, a claim that requires process evidence precisely because it is about the training-time origin of a behavior.

What remains open. Under the stratified view: induction heads appear to be point-localized objects, but whether the relevant causal variable is localized to a one-dimensional subspace (object stratum) or spans a small-dimensional subspace is not established. Under the structural view: the holonomy of the induction head transport map has not been characterized. Under the subspace view: the exact subspace of the residual stream encoding the “previous occurrence” variable and the “current token” variable has not been jointly identified via DAS.

Verdicts. *Object view:* established across model families. *Role view:* confirmed—one of the few cases where role identity is genuinely operationalized and independently tested. *Process view:* established via checkpoint analysis and composition score evolution. *Subspace and structural views:* not attempted. The convergence across object, role, and process views is why the induction head result can support relatively strong claims—it is the existence proof that the framework’s triangulation requirement is satisfiable. The gaps in subspace and structural evidence are where further work would strengthen or qualify those claims.

5.3 Superposition

The hypothesis that neural networks represent more features than they have dimensions—*superposition*—has become one of the central organizing problems in mechanistic interpretability (Elhage et al., 2022). Substantial engineering effort has been directed at “solving” superposition, primarily through sparse autoencoders (SAEs) that project activations into higher-dimensional spaces where features are disentangled (Bricken et al., 2023; Templeton et al., 2024). But whether superposition is a *problem to be solved* depends on which mechanistic view one adopts. The framework reveals that the field’s framing presupposes an ontological commitment that is rarely stated and never defended.

Object view. Under the Object view, features are discrete, countable parts of the model’s representation. Superposition occurs when these parts are “compressed” into fewer dimensions than there are features, entangling them in the activation space. SAEs exist precisely to undo this compression: by training a wide autoencoder with a sparsity penalty, one recovers a dictionary of monosemantic units, each corresponding to a single feature (Bricken et al., 2023). The implicit assumption is that a clean decomposition exists underneath the entanglement.

This framing encounters a difficulty it lacks the internal resources to resolve. Different SAE widths yield different feature sets: a 4×-width SAE finds features that a 16×-width SAE splits further, and a 64×-width SAE splits further still (Templeton et al., 2024). Which width gives the “right” features? The Object view requires a determinate inventory of parts, but the inventory changes with the magnifying glass. No principled criterion within the view selects a canonical width.

Subspace view. The Subspace view reframes features as subspaces rather than individual directions. Superposition is not a compression artifact but a geometric property: subspaces can overlap, and in high-dimensional spaces, approximate orthogonality is abundant enough that substantial overlap is expected rather than pathological. The question shifts from “how do we disentangle features?” to “how much do representational subspaces interfere, and does that interference degrade downstream computation?”

This reframing makes the SAE width problem dissolve into a measurable quantity. Different SAE widths correspond to different granularities of subspace decomposition, and the relationship between them can be characterized by Grassmannian distances between the recovered subspaces across widths and random seeds. If subspaces converge, they identify robust geometric structure; if they fragment, the decomposition at that granularity is an artifact.

Structural view. The Structural view offers the deepest reframe. If mechanistic structure is constituted by computational relationships rather than any particular parameterization, then superposition is a *gauge choice*: a consequence of the coordinate system in which we inspect the model, not an intrinsic property of the computation. Two models with different superposition patterns may implement identical input-output mappings and preserve identical computational invariants. What matters structurally is not the degree of superposition but what computation is invariant under changes of basis.

Perspectival view. The claim “neural networks are in superposition” presupposes that there exist discrete features being compressed into fewer dimensions—but this is the Object view’s ontology imposed as a universal empirical assumption. The canonical demonstration of superposition (Elhage et al., 2022) uses toy models trained to reconstruct sparse, known input features. In that setting, superposition is rigorously defined because the ground-truth features are specified by construction. But large language models were not trained to reconstruct a sparse feature dictionary; they were trained to predict tokens. The mapping from the toy setting to real networks requires the assumption that LM representations decompose into sparse discrete features in the first place—precisely the assumption that is under investigation.

The Perspectival view flags this as a circularity risk: SAEs are trained with a sparsity prior that *enforces* decomposition into discrete units, then the resulting dictionaries are taken as evidence that discrete features exist (Leask et al., 2025). This does not mean SAEs are uninformative, but it means the evidence they provide for the superposition hypothesis is weaker than typically assumed.

Verdicts. *Object view*: superposition is real and must be solved, but the decomposition is *underdetermined*—no principled criterion selects the correct SAE width. *Subspace view*: superposition is *testable geometry*—measure Grassmannian distances across widths and seeds to determine which subspaces are robust. *Structural view*: superposition is a *gauge choice*—the productive question is what computation is invariant under reparameterization. *Perspectival view*: the claim that models “are in superposition” is *theory-laden*—it imports the Object view’s ontology as an unstated assumption, supported by toy models whose training objective differs from that of real language models. The field’s substantial investment in “solving superposition” is coherent under the Object view, but it assumes that view is correct—an assumption that is rarely stated and never defended.

6 Six Decision Points

Six recurring forks in empirical interpretability work change in meaning depending on which view is operative. We examine what each view implies for each fork and, where views make conflicting predictions, what evidence would adjudicate.

6.1 Localized vs. Distributed

The question is whether a mechanism is implemented by a small, specific set of components or spread across many.

Under the *object view*, this is an empirical question: which minimal component set is causally necessary and sufficient? Under the *subspace view*, it is about the dimensionality and localizability of the causal subspace: a low-rank subspace localized in a small number of attention heads is one answer; a high-dimensional subspace spanning many heads is another. Under the *stratified view*, it is a question about which stratum the evidence supports.

The views come apart in practice. Activation patching may identify a small object-level component set as causally necessary, while DAS identifies a high-dimensional subspace spanning many components. These need not be contradictory: the patching identifies the components whose activation values lie in the relevant subspace; DAS characterizes the subspace. The stratified view provides the language to say that the two results are compatible descriptions at different levels of resolution.

6.2 Object vs. Role

The question is whether a mechanism claim is about a specific component or about a functional role that component plays. Under the *object view*, the claim is about the component; under the *role view*, it is about the function.

The distinction matters for generalization. An object claim does not entail that any component in any other model is the same mechanism. A role claim does—if the role is operationalized and tested. A common pattern is to establish object identity and then introduce a role label in the same passage (“head 9.9, which we call a name mover, ...”): the object identity is established, and the role label is natural shorthand, but it carries additional commitments that the object evidence alone does not discharge.

6.3 Single vs. Triangulated Evidence

The question is whether a mechanistic claim requires evidence from a single method or convergent evidence from multiple methods with independent assumptions.

Under the *object view* with activation patching, a single strong patching result may be sufficient. Under the *subspace view*, weight-space and activation-space convergence strengthens the claim (both should identify the same subspace if the view is correct). Under the *perspectival view*, single-method evidence is always vulnerable to that method’s confounds: triangulation is definitionally required.

The case for triangulation is not merely a caution. If a method m can only support claims in its range $C(m)$, and the target claim is not in $C(m)$, then no amount of evidence from m can support the claim—however strong that evidence is.

6.4 Static vs. Process

The question is whether the mechanism is a final-checkpoint artifact or a formation process. This is a clean view-level distinction with a clean evidential consequence: process claims need process evidence; checkpoint claims cannot support process conclusions.

This matters most for: (a) training-dynamics hypotheses (“the model learned to do X because ...”), (b) phase-transition-like generalization patterns, and (c) claims about the origin of in-context learning behaviors. In all three cases, checkpoint evidence establishes what the mechanism is at convergence but not how it came to be.

6.5 Subspace vs. Structural

The question is whether mechanism identity is determined by the subspace a mechanism occupies or by the gauge-invariant structure of the computation it performs. A subspace claim is basis-invariant: the subspace $\text{span}(Q)$ is the same regardless of which orthonormal basis of Q is used. But it is not symmetry-invariant: two subspace mechanisms in the same gauge orbit (related by a computation-preserving symmetry that rotates the subspace) would be the same mechanism under the structural view but potentially different under the subspace view.

The structural view is more realist and harder to operationalize. The subspace view is more tractable and directly supported by current DAS methodology. For most practical questions in the field, the subspace view is appropriate; the structural view becomes important when comparing across parameterizations of the same architecture or when making claims that should be invariant to training randomness.

6.6 Behavioral vs. Mechanistic

The question is whether a finding is about a model’s *behavioral propensity* (how it tends to behave on a distribution) or about a *mechanism* (an internal structure that produces behavior). Steinhardt (2026) argues—in the context of evaluation methodology, not ontology—that the field over-invests in capability evaluation (can the model do X ?) and under-invests in propensity evaluation (does the model typically

do X , and under what conditions?). The distinction is relevant here because different views assign different status to behavioral findings.

Under the *instrumental view*, a behavioral finding—the model reliably outputs Y given X —is sufficient. A propensity is what you care about, and a mechanism is irrelevant. Under the *object view*, a behavioral finding is a starting point: the question becomes which components produce that behavior, and whether they are necessary. A capability finding (the model *can* do X) is weaker than a propensity finding (the model *typically does* X) because the former requires only that the relevant mechanism exists, while the latter requires that it be routinely recruited.

The distinction matters most for safety. A steering vector that suppresses refusal behavior on a test distribution establishes an instrumental capability—not a mechanistic understanding of refusal (Turner et al., 2024). Whether that vector *is* the refusal mechanism, or merely interferes with it, requires at least object-view evidence (ablation of specific components) and ideally subspace-view evidence (stability of the intervention direction across distributions). The gap between instrumental evidence and the safety conclusions drawn from it is the central evidence deficit identified by the open-problems analysis in Appendix D.

7 Using the Framework

7.1 The View Declaration Template

The primary practical implication of this paper is that mechanistic interpretability claims should state the view under which they are made. We propose a five-line *view declaration* that a paper would include in its methods section:

View declaration. *Ontology*: [what kind of entity counts as a mechanism]. *Identity*: [when two descriptions refer to the same mechanism]. *Evidence*: [what measurements support the claim]. *Formalism*: [what mathematical language is used]. *Target*: [what phenomenon is being explained].

For example, the IOI circuit paper (Wang et al., 2023) would declare:

View declaration. *Ontology*: concrete attention heads (Object view). *Identity*: component overlap within GPT-2 Small. *Evidence*: activation patching, path patching, mean ablation. *Formalism*: directed graph over 26 heads. *Target*: indirect object identification on the IOI distribution.

This is not bureaucratic overhead—it is a precision requirement of the same kind as reporting which distribution patching was run on. Stating the view makes evidence-claim mismatches immediately visible: if a paper declares Object ontology but draws Role conclusions (“head 9.9 is a name mover”), the gap between the declared evidence and the stated claim becomes a checkable coherence condition rather than an unstated assumption.

7.2 Common Coherence Violations

Table 3 lists the five most common patterns in which the evidence type and the claim type are mismatched. Each row diagnoses the violation and identifies the evidence that would resolve it.

Violation	What it looks like	What would fix it
Object evidence, Role claim	“Head X is a name mover” (ablation only)	Role-specific prediction test on held-out constructions
Unconstrained DAS	“87% IIA” without specifying alignment class	State alignment class; rerun with linear constraint (Sutter et al., 2025)
Object claim, Structural conclusion	“Circuit is robust to fine-tuning”	Reparameterization stability test
Process claim, no checkpoint evidence	“Mechanism formed via phase transition”	Report composition score evolution across checkpoints
Contrastive mechanism stated absolutely	“The bias circuit encodes gender”	Report foil set; vary foil systematically

Table 3: Five common coherence violations and their resolutions.

7.3 Independent Convergence

Williams et al. (2025) argue from philosophy of science that MI needs philosophical foundations—for clarifying concepts, refining methods, and navigating epistemic complexity. Their three central examples (decomposition identity, feature ontology, deception detection) map directly onto view-level analyses in this paper: “no single correct decomposition” corresponds to the object-subspace distinction (Section 3.3); “vehicle vs. content” corresponds to the object-role split applied to features; and “behavioral detection is insufficient for deception” corresponds to the instrumental view’s ceiling (Appendix D). The convergence from independent starting points—philosophy of science vs. measurement theory—strengthens both analyses.

7.4 Views as Research Programs

The object and role views primarily organize existing empirical practice. The subspace, structural, and stratified views generate positive research programs: each calls for methods, results, and formalism that do not yet exist. For the subspace view: systematic DAS studies with stability tests across models and distributions. For the structural view: holonomy measurements, gauge-invariance checks, and the development of geometric formalism. For the stratified view: dimensionality diagnostics and stratum-transition tests. The atlas is therefore not merely a taxonomy of current practice; it is a map of what is missing.

8 Cross-View Promotion

Promotion conditions. A common question is: under what conditions can a result at one view be promoted to a higher-commitment view? The following are preliminary promotion conditions:

- **Object $\rightarrow$ Role.** The component must be tested with an independently operationalized role specification on held-out constructions. If the role predicts behavior beyond the original discovery distribution, the result can be stated as a role claim.
- **Role $\rightarrow$ Subspace.** The functional role must be shown to be mediated by a specific low-dimensional subspace (e.g., via linear DAS with linear alignment constraints), and this subspace must be stable across prompt distributions with small geodesic distance on $\text{Gr}(k, d)$.
- **Subspace $\rightarrow$ Structural.** The subspace must survive computation-preserving symmetries of the weight space. If the subspace recovered by DAS changes under a reparameterization that does not change the model’s computation, the subspace claim is not structural. Holonomy measurements provide a gauge-invariant fingerprint.
- **Any $\rightarrow$ Process.** The static claim must be supplemented with training-dynamics evidence: checkpoint analysis showing when and how the mechanism formed, formation knockouts, or loss-curve structure. No amount of final-checkpoint evidence can substitute.

Each promotion requires new evidence of the kind specified by the target view. Evidence from the source view does not accumulate toward the target view’s standard—it answers a different question. The chain is not sequential: promotion can skip levels when the evidence supports it. For example, an Object-level result can promote directly to Subspace if linear DAS identifies a causal subspace spanning the relevant components, without requiring an intermediate Role specification. The subspace evidence subsumes the object evidence (the components are where the subspace has support) but does not require role-level mediation.

View selection: a practical heuristic. The framework is pluralist, but a researcher facing a concrete question needs guidance on which view to adopt. The following heuristic matches the scope of the claim to the minimum view that can support it:

- If the claim is about *a specific model on a specific distribution*, the **Object view** is sufficient.
- If the claim should *generalize across models* (“Pythia and GPT-2 both have this mechanism”), at least the **Role view** is required—component indices are trivially not cross-model.
- If the claim involves a *distributed representation* not localized to a single component, the **Subspace view** is needed.
- If the claim should *survive reparameterization* (permutation, rescaling, rotation of weight matrices), the **Structural view** is needed.
- If the claim is about *how the mechanism formed* or why it appeared at a specific training step, the **Process view** is needed—no amount of checkpoint evidence can substitute.
- If the mechanism’s *apparent type changes with measurement resolution*, the **Stratified view** is the appropriate language.

The heuristic is conservative: it recommends the lowest-commitment view that matches the claim’s scope. Higher-commitment views are always available but require correspondingly stronger evidence.

Connections to neighboring fields. The five-axis framework has natural translations into several disciplines whose concepts it draws on.

Philosophy of science. The ontology axis corresponds to the central question of the new mechanist literature: what counts as a mechanism (Machamer et al., 2000; Craver, 2007; Illari & Williamson, 2012; Glennan, 1996)? The identity axis operationalizes the question that Lewis’s account of functional identification (Lewis, 1972) raises: under what conditions does a mechanism remain “the same”? The evidence axis inherits the eliminativist structure of Earman’s Bayesian epistemology (Earman, 1992) and the domain-triangulation principle of Howson and Urbach (Howson & Urbach, 1989): independent evidence types eliminate distinct confounders. Massimi’s perspectival realism (Massimi, 2022) maps directly onto the perspectival view, with the important distinction that perspectival evidence does not entail perspectival truth.

Physics. The structural view’s gauge-invariance requirement is the direct analogue of gauge symmetry in field theory: two weight configurations related by a computation-preserving transformation are physically the same mechanism, just as two vector potentials related by a gauge transformation describe the same electromagnetic field (Nakahara, 2003). The stratified view borrows the renormalization group’s central insight (Wilson, 1971; Goldenfeld, 1992): effective descriptions change with measurement scale, and the interesting objects are the fixed points (stable strata) and relevant operators (structure that survives coarse-graining). Universality classes in physics correspond to mechanisms that appear identical at a given resolution despite differing microscopically.

Mathematics. The formalism axis is not merely notational. The choice of mathematical structure constrains what can be stated precisely and what can be proved. Grassmannian geometry (Nakahara, 2003) makes subspace proximity a metric statement; fiber bundles make gauge invariance a theorem rather than a heuristic check; Whitney stratification (Whitney, 1965; Golubitsky & Guillemin, 1973) gives a rigorous account of

how mechanism identity changes across strata boundaries. Each formalism carries proof obligations that the corresponding informal claim does not.

Cognitive science and neuroscience. The localization–distribution spectrum (object view through stratified view) mirrors the longstanding debate between localizationism and distributed coding in systems neuroscience (Bechtel & Richardson, 1993). The double dissociation methodology in neuropsychology—showing two functions can be selectively impaired independently—is the nearest analogue to the necessity and sufficiency tests used in the object view. Marr’s tri-level account (Marr, 1982) organizes explanatory targets; the five axes here organize ontological commitments. Both are needed, and they are complementary: two researchers at the same Marr level can hold different mechanistic views.

9 Open Questions

Several questions are raised by the framework and not resolved:

1. *Process identity.* The identity criterion for the process view (same dynamical basin or trajectory type) is underspecified relative to the object, subspace, and structural criteria. Formalizing process identity is an open problem.
2. *Cross-view promotion.* Under what conditions can an object-level result be promoted to a subspace-level or structural-level result? Section 8 gives preliminary conditions, but a systematic account of inter-view inference is missing.
3. *Completeness of the atlas.* The nine views are identified by reading the literature, not derived from the five axes. Whether there are important views not represented—a Marr computational-level view (Marr, 1982), a representational-content view, a population-level statistical view—is an open question.
4. *Sufficiency of the five axes.* The five axes may be too many or too few. Evidence and formalism are tightly constrained by ontology and identity (the determination chain), raising the question of whether they are genuinely free axes or downstream consequences. Conversely, the framework omits a normative dimension—it diagnoses coherence but not adequacy for a given purpose. Whether adequacy is a sixth axis or a meta-level concern about the framework is open.
5. *View-independence of the framework.* The coherence conditions (§3.2) assume crisp equivalence relations. Both the Perspectival view (coherence across perspectives) and the Process view (dynamical basin membership) have inherently fuzzy identity criteria that may not satisfy transitivity. Whether this is a limitation of the views or of the framework’s formalization is unresolved.
6. *Views and safety-relevant inference.* Safety applications (monitoring, alignment auditing) require mechanistic claims that are robust to adversarial distribution shift. What minimum view supports safety-relevant inference? The object view is likely insufficient—a head identified as safety-relevant on one distribution may not generalize. The structural or stratified views may be necessary, but their evidential demands are currently impractical at scale.
7. *Implicit view commitments of existing programs.* The monosemanticity program (Bricken et al., 2023; Templeton et al., 2024) implicitly assumes that individual SAE dictionary elements are the right ontological unit (object view) with feature-level identity. The causal abstraction program (Geiger et al., 2021; 2024) assumes role-level identity via interchange intervention. Making these commitments explicit would clarify what each program can and cannot establish.

10 Conclusion

Mechanistic interpretability is not only a search for circuits. It is a search for the right kind of object to call a mechanism. The field already uses multiple mechanistic views, typically implicitly, and making those views explicit improves both the precision of claims and the appropriateness of evidence.

The central contribution is a taxonomy: every mechanistic interpretability result assumes a mechanistic view—a bundle of commitments about ontology, identity, evidence, formalism, and target—whether or not it is stated. Coherence between these components is not automatic (§3.2), and the determination chain—ontology constrains identity constrains formalism—makes mismatches between them diagnosable.

The atlas of nine views—object, role, subspace, structural, process, stratified, perspectival, instrumental, contrastive—covers the main options currently in use or emerging. Each view is appropriate for some questions and inappropriate for others. The paper is pluralist: the goal is not to collapse this plurality into a master ontology but to make view choice explicit, justifiable, and revisable. The induction head result remains the canonical illustration of what convergence across views looks like: object, role, and process views all point to the same entity, which is why the result supports claims that single-view results cannot.

This paper addresses the ontological prerequisites for mechanistic claims. Companion work addresses the validity structure of individual claims (Tower, 2026a), the reference conditions for cross-model identity (Tower, 2026b), and the composition conditions under which validated claims aggregate into system-level understanding (Tower, 2026c).

References

- Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Rimsky, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. *arXiv:2406.11717*, 2024.
- Mikita Balesni, Marius Hobbhahn, David Lindner, Alexander Meinke, Tomek Korbak, Joshua Clymer, Buck Shlegeris, Jeremy Scheurer, Charlotte Stix, et al. Towards evaluations-based safety cases for AI scheming. *arXiv:2411.03336*, 2024.
- Karan Bali, Jack Stanley, Praneet Suresh, and Danilo Bzdok. Quantifying LLM attention-head stability: Implications for circuit universality. *arXiv:2602.16740*, 2026.
- William Bechtel and Robert C. Richardson. *Discovering Complexity: Decomposition and Localization as Strategies in Scientific Research*. Princeton University Press, Princeton, NJ, 1993.
- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022.
- Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, and Adam Kalai. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2016.
- Trenton Bricken, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nicholas L. Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Zac Hatfield-Dodds, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E. Burke, Tristan Hume, Shan Carter, Tom Henighan, and Chris Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemanticity>.
- Stephen Casper, Jason Lin, Joe Kwon, Gilbert Tok, and Dylan Hadfield-Menell. Black-box access is insufficient for rigorous AI audits. *arXiv:2401.14446*, 2024.
- Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Carl F. Craver. *Explaining the Brain: Mechanisms and the Mosaic Unity of Neuroscience*. Oxford University Press, Oxford, 2007.
- Lindley Darden. *Reasoning in Biological Discoveries: Essays on Mechanisms, Interfield Relations, and Anomaly Resolution*. Cambridge University Press, Cambridge, 2006.

-
- John Earman. *Bayes or Bust? A Critical Examination of Bayesian Confirmation Theory*. MIT Press, Cambridge, MA, 1992.
- Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021. <https://transformer-circuits.pub/2021/framework>.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, et al. Toy models of superposition. *Transformer Circuits Thread*, 2022.
- Guilherme Franco, Lucas M. Tassis, Achille Rohr, and Mark Crovella. Finding highly interpretable prompt-specific circuits in language models. arXiv:2602.13483, 2026.
- Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Atticus Geiger, Zhengxuan Wu, Christopher Potts, Thomas Icard, and Noah D. Goodman. Finding alignments between interpretable causal variables and distributed neural representations. In *Conference on Causal Learning and Reasoning (CLEaR)*, 2024.
- Stuart Glennan. Mechanisms and the nature of causation. *Erkenntnis*, 44(1):49–71, 1996.
- Nigel Goldenfeld. *Lectures on Phase Transitions and the Renormalization Group*. Addison-Wesley, Reading, MA, 1992.
- Martin Golubitsky and Victor Guillemin. *Stable Mappings and their Singularities*. Springer-Verlag, New York, 1973.
- John Hewitt and Christopher D. Manning. A structural probe for finding syntax in word representations. In *Proceedings of NAACL-HLT*, 2019.
- Colin Howson and Peter Urbach. *Scientific Reasoning: The Bayesian Approach*. Open Court, La Salle, IL, 1989.
- Evan Hubinger, Chris van Merwijk, Vladimir Mikulik, Joar Skalse, and Scott Garrabrant. Risks from learned optimization in advanced machine learning systems. arXiv:1906.01820, 2019.
- Phyllis McKay Illari and Jon Williamson. What is a mechanism? Thinking about mechanisms across the sciences. *European Journal for Philosophy of Science*, 2(1):119–135, 2012.
- Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, et al. Measuring faithfulness in chain-of-thought reasoning. arXiv:2307.13702, 2023.
- Patrick Leask, Joshua Mendel, Maria Koumeri, Nenad Tomasev, Kenneth Li, and Wes Gurnee. Sparse autoencoders do not find canonical units of computation. In *International Conference on Learning Representations (ICLR)*, 2025.
- David Lewis. Psychophysical and theoretical identifications. *Australasian Journal of Philosophy*, 50(3):249–258, 1972.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

-
- Kenneth Li, Aspen K. Hopkins, David Bau, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Emergent world representations: Exploring a sequence model trained on a synthetic task. In *International Conference on Learning Representations (ICLR)*, 2023.
- Peter Lipton. Contrastive explanation. *Royal Institute of Philosophy Supplements*, 27:247–266, 1990.
- Peter Machamer, Lindley Darden, and Carl F. Craver. Thinking about mechanisms. *Philosophy of Science*, 67(1):1–25, 2000.
- David Marr. *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information*. W. H. Freeman, San Francisco, 1982.
- Michela Massimi. *Perspectival Realism*. Oxford University Press, Oxford, 2022.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Jack Merullo, Carsten Eickhoff, and Ellie Pavlick. Circuit component reuse across tasks in transformer language models. In *International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.08744.
- Aaron Mueller, Atticus Geiger, Stefan Heimersheim, Mor Geva, and others. Mechanistic interpretability benchmark: A comprehensive evaluation for circuit discovery methods. 2025. <https://github.com/mechanistic-interpretability-benchmark/mib>.
- Mikio Nakahara. *Geometry, Topology and Physics*. Institute of Physics Publishing, Bristol, 2nd edition, 2003.
- Neel Nanda. 200 concrete open problems in mechanistic interpretability. Blog post, 2022. <https://www.neelnanda.io/mechanistic-interpretability/200-open-problems>.
- Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, and Jacob Steinhardt. Progress measures for grokking via mechanistic interpretability. In *International Conference on Learning Representations (ICLR)*, 2023.
- Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022. <https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads>.
- Alethea Power, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra. Grokking: Generalization beyond overfitting on small algorithmic datasets. arXiv:2201.02177, 2022.
- Willard Van Orman Quine. On what there is. *Review of Metaphysics*, 2(5):21–38, 1948.
- Schmidt Futures. Trustworthy AI research agenda. Request for proposals, 2026.
- Lee Sharkey, Bilal Chughtai, Joshua Batson, Jack Lindsey, Jeff Wu, Lucius Bushnaq, Nicholas Goldowsky-Dill, Stefan Heimersheim, et al. Open problems in mechanistic interpretability. arXiv:2501.16496, 2025.
- Jacob Steinhardt. The case for evaluating model behaviors. Alignment Forum, May 2026.
- Alan Sun and Mariya Toneva. Tracking equivalent mechanistic interpretations across neural networks. In *International Conference on Learning Representations (ICLR)*, 2026.
- Denis Sutter, Julian Minder, Thomas Hofmann, and Tiago Pimentel. The non-linear representation dilemma: Is causal abstraction enough for mechanistic interpretability? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Aaquib Syed, Can Rager, and Arthur Conmy. Attribution patching outperforms automated circuit discovery. arXiv:2310.10348, 2023.

-
- Adly Templeton et al. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. *Anthropic Transformer Circuits Thread*, 2024. <https://transformer-circuits.pub/2024/scaling-monosemanticity>.
- Curt Tigges, Oskar John Hollinsworth, Atticus Geiger, and Neel Nanda. LLM circuit analyses are consistent across training and scale. arXiv:2407.10827, 2024.
- Elliot Tower. Mechanistic Validity: Evidence Standards for Mechanism Claims. arXiv preprint, 2026.
- Elliot Tower. Mechanistic reference: When does a mechanism term pick out the same thing? arXiv preprint, 2026.
- Elliot Tower. When do circuit discoveries compose into understanding? arXiv preprint, 2026.
- Alex Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization. arXiv:2308.10248, 2024.
- Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- Bas C. van Fraassen. *The Scientific Image*. Oxford University Press, Oxford, 1980.
- Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Kevin Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: A circuit for indirect object identification in GPT-2 Small. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2211.00593.
- Junxuan Wang, Xuyang Ge, Wentao Shu, Qiong Tang, Yunhua Zhou, Zhengfu He, and Xipeng Qiu. Towards universality: Studying mechanistic similarity across language model architectures. arXiv:2410.06672, 2024.
- Hassler Whitney. Tangents to an analytic variety. *Annals of Mathematics*, 81(3):496–549, 1965.
- Iwan Williams, Ninell Oldenburg, Ruchira Dhar, Joshua Hatherley, Constanza Fierro, Nina Rajcic, Sandrine R. Schiller, Filippos Stamatiou, and Anders Søgaard. Mechanistic interpretability needs philosophy. arXiv:2506.18852, 2025.
- Kenneth G. Wilson. Renormalization group and critical phenomena. I. Renormalization group and the Kadanoff scaling picture. *Physical Review B*, 4(9):3174–3183, 1971.

A Coherence Violations: A Diagnostic Checklist

The following patterns indicate a coherence violation in the view underlying a claim.

- (1) **Ontology–identity mismatch.** The ontology declares mechanisms to be concrete components, but the identity criterion is functional (two different components can be the same mechanism). Resolution: adopt the role view, which has functional identity built in, or restate the claim as object-level and drop cross-model generalization.
- (2) **Evidence out of range.** A structural or subspace claim is supported only by activation-patching evidence. Resolution: identify the view that patching evidence actually supports (object view) and state the structural or subspace claim as a hypothesis requiring further evidence.
- (3) **Unrestricted alignment.** A subspace claim is supported by DAS/IIA without specifying the alignment class or reporting subspace stability. By the Sutter result (Sutter et al., 2025), high IIA under unrestricted nonlinear alignment is consistent with any causal structure. Resolution: restrict the alignment class to linear maps or transport-respecting maps and report the eigenvalue gap.
- (4) **Process claim from checkpoint evidence.** A claim about how a mechanism formed or why a behavior emerged at a certain scale is supported only by final-checkpoint analysis. Resolution: either restate as a static claim or obtain training-dynamics evidence.
- (5) **Object-to-role slide.** A component is identified by ablation and immediately labeled with a role without an independent test of the role specification. Resolution: operationalize the role prediction and test it on held-out constructions.

B Extended Case Studies

Each case study applies the view taxonomy from §3 to a concrete phenomenon in mechanistic interpretability. The goal is not to resolve the phenomenon but to show how the choice of view determines what counts as explanation, what evidence is relevant, and what conclusions are licensed. Cases are grouped by type: phenomena (§B.1–B.6), debates dissolved (§B.7–B.11), safety applications (§B.12–B.13), and methodological foundations (§B.14).

B.1 SAE Features: Decomposition or Projection?

Sparse autoencoders decompose activations into overcomplete dictionaries of interpretable directions (Bricken et al., 2023). However, the features recovered depend on dictionary width, random seed, and architectural choices (Leask et al., 2025). Whether this variability is a problem or an expected property depends entirely on the view.

Object view. Each dictionary element is a candidate mechanistic primitive—a one-dimensional object in activation space. Different SAE widths recovering different feature sets is troubling: Bricken et al. (2023) demonstrate progressive feature splitting as width increases. If the model possesses ground-truth features, which width recovers them? The Object view provides no principled selection criterion—every feature splits further at higher widths with no evident convergence to indivisible units.

Subspace view. Individual features are basis vectors; the causally relevant structure is the subspace they collectively span. Different widths producing different basis vectors for overlapping subspaces is expected—the geometric analogue of choosing different coordinate systems. The empirical test is direct: measure Grassmannian distance between feature subspaces across widths and seeds. If distances are small, SAEs recover real geometric structure regardless of which basis they select. Feature splitting becomes basis refinement within a stable subspace, not ontological fragmentation.

Perspectival view. SAE features are projections of activations through sparse dictionary learning—a specific methodological perspective with its own inductive biases. A different method (NMF, ICA, clustering) projects different features. Neither is wrong; neither is uniquely “true.” SAEs trained on random activations still produce features that receive plausible labels from automated interpretability pipelines—exactly what the Perspectival view predicts. Leask et al. (2025) demonstrate that different training runs produce different features, consistent with features being partly artifacts of the measurement procedure.

Feature absorption. A safety-relevant sub-case: a narrow SAE may isolate a feature used for monitoring; at higher widths, this feature is absorbed into a more general feature. Under the Object view, this is alarming—a monitored mechanism has vanished. Under the Subspace view, the question is whether the subspace remains detectable even as the basis rotates. If the Grassmannian point is stable under width perturbation, monitoring can target the subspace rather than any particular basis vector.

Verdicts. *Object:* weak—steering demonstrates causal relevance but different runs disagree on which features exist. *Subspace:* testable—Grassmannian convergence across widths and seeds is directly measurable. *Perspectival:* critical—discriminant validity against random-activation baselines is required before any feature can be called “real” rather than projected.

B.2 Gender Bias Circuits

Several studies identify specific attention heads as causally involved in gendered pronoun resolution and propose ablating them to remove stereotypical gender bias (Bolukbasi et al., 2016; Vig et al., 2020). The difficulty is that the same heads process both stereotypical bias *and* grammatical gender agreement, so the safety conclusion—whether bias can be surgically removed—depends on which view frames the analysis.

Object view. Activation patching identifies a small set of attention heads whose ablation degrades gendered pronoun prediction. The object-level claim is well-supported: specific components are implicated by causal intervention. However, the object view treats each head as an atomic unit and cannot distinguish whether a single head participates in one mechanism or two.

Contrastive view. This is the decisive view. The circuit identified as the “gender bias circuit” is the set of components whose behavior *differs* between stereotypical and counter-stereotypical completions (“The nurse said *she*” vs. “The nurse said *he*”). But a different foil—gendered vs. gender-neutral completions—identifies a different circuit that includes grammatical gender processing. If the foil is counter-stereotypical, the circuit isolates what is specific to bias; if the foil is gender-neutral, the circuit includes the shared grammatical machinery. The safety conclusion “ablating the bias circuit does not degrade capability” is valid only under the first foil. Under the second, ablation necessarily degrades grammatical gender agreement. The mechanism *is* foil-relative, and any intervention justified by a contrastive analysis inherits the scope of the foil that defined it.

Role view. Is the functional role “gender bias head” or “gender agreement head”? The label depends on which behavior operationalizes the role. A head that mediates information from gendered nouns to pronoun positions plays the gender-agreement role on grammatically unambiguous sentences and the bias-amplification role on occupational stereotypes. Without independent role tests across multiple syntactic constructions, the functional label is asserted, not confirmed.

Subspace view. Do stereotypical bias and grammatical gender occupy *distinct* linear subspaces? If so, a DAS-style rotation could isolate the bias subspace and enable surgical intervention at the level of directions rather than heads (Li et al., 2023a). If bias and grammar share a subspace, no linear intervention can remove one without affecting the other. No published DAS analysis has trained separate bias and grammar probes and measured the principal angle between their subspaces.

Verdicts. *Object:* established—specific heads identified by causal intervention. *Contrastive:* critical—the foil determines which circuit is identified and whether the safety conclusion holds. *Role:* contested—

“bias head” vs. “gender agreement head” depends on operationalization, not on the head itself. *Subspace*: untested—no DAS analysis separates the bias subspace from the grammatical-gender subspace.

B.3 Factual Recall and Knowledge Editing

Meng et al. (2022) localize factual knowledge to specific MLP layers via causal tracing—corrupting the subject token and restoring individual layers—then edit facts by rank-one updates to those weights (ROME). The localization is object-view evidence; the editing is an instrumental intervention. The gap between them is where related facts break.

Object view. Causal tracing identifies early-site MLP layers as causally necessary for recalling a specific fact: restoring only those layers after subject corruption recovers the correct completion. The object claim is valid within the test distribution. The limitation is that the object view treats each fact as independent—it asks “where is this fact stored?” but does not ask whether modifying that location will have side effects on causally entangled facts.

Instrumental view. ROME edits succeed: after a rank-one update, the model outputs the desired new fact. This is instrumental evidence—the intervention achieves the desired input-output behavior. But instrumental sufficiency does not imply mechanistic understanding. The edit works in the same sense that setting a variable to a constant “works”: it overwrites the output without necessarily operating through the mechanism that originally produced it.

Structural view. Editing “The Eiffel Tower is in Paris” to “...Rome” sometimes breaks “The capital of France is ...” This failure demonstrates that the object-level localization does not capture the relational structure connecting associated facts. The mechanism is not a single updatable slot but a distributed structure where facts mutually constrain each other through shared weight geometry. A rank-one edit modifies one entry while leaving the constraints intact, producing inconsistency.

Subspace view. The causal variable for factual recall may occupy a multi-dimensional subspace. If the relevant information spans a k -dimensional subspace with $k > 1$, then ROME’s rank-one update is incomplete by construction: it modifies one direction but leaves $k - 1$ directions carrying residual information about the original fact. No published work has measured the effective dimensionality of the factual recall subspace.

Verdicts. *Object*: established—causal tracing provides valid localization. *Instrumental*: demonstrated—edits achieve the desired output without mechanistic understanding. *Structural*: violated—related-fact failures reveal relational structure that the object-level localization missed. *Subspace*: untested—the dimensionality of the factual-recall subspace has not been measured.

B.4 Hallucination

Language models generate confident, factually incorrect outputs. Two mechanistically distinct failure modes are worth separating: knowledge absence (the correct fact was never encoded) and retrieval failure (the correct fact is encoded but the wrong candidate wins the competition for the output).

Object view. Activation patching from a “correct-answer” forward pass into a “hallucinating” forward pass can recover the correct completion, localizing the failure to specific mid-to-late layers. However, if the correct and incorrect candidates share computational components, the object view cannot cleanly localize the failure to a specific component that “causes” hallucination. The pathology is not a broken part but a competition between parts functioning as designed.

Role view. The mechanism is the fact-retrieval role: a functional capacity distributed across multiple components. Hallucination is role competition—the retrieval role and a distributional-prior role (which promotes high-frequency completions regardless of context) both activate, and the latter dominates. This framing predicts that the same components may play different roles on different prompts.

Subspace view. The causal variable is “which entity is currently promoted as the answer candidate.” In the hallucinating case, the wrong entity has a higher projection onto the answer-readout subspace. Systematic DAS analysis with IIA for factual-recall hallucination has not been published.

Structural view. Hypothesis: facts encoded with lower effective-rank weight structure hallucinate more frequently under retrieval competition. The intuition is that low-effective-rank storage creates a robust attractor for the stored fact, while high-effective-rank storage is more susceptible to interference. This hypothesis has not been tested by measuring the effective rank of fact-specific weight structure.

Verdicts. *Object:* Tier 2—patching localizes recall to mid-to-late layers but cannot isolate the hallucination-specific failure. *Role:* Tier 2—the role-competition hypothesis is consistent with evidence but lacks direct causal tests. *Subspace:* Tier 1—no DAS validation of the answer-candidate subspace. *Structural:* Tier 1—the effective-rank hypothesis is theoretically motivated but empirically untested.

B.5 Grokking

A model suddenly generalizes long after memorizing its training set (Power et al., 2022; Nanda et al., 2023). The mechanistically interesting phenomenon is the *transition*—no single-checkpoint analysis can capture what makes grokking distinctive.

Process view. The mechanism *is* the formation trajectory: a phase transition from a memorization circuit to a generalization circuit. The identity of the grokking mechanism is constituted by this trajectory—by the sequence of weight-space changes and the checkpoint-by-checkpoint emergence of the algorithmic solution. No static description of the final circuit explains *why* generalization occurred, *when* it occurred, or what determined the phase-transition point.

Object view. After grokking, the circuit implementing the generalized computation can be identified. This object-level description is valid—it tells us *what* the model computes after the transition. But it cannot distinguish a model that grokked from one trained directly on the generalization objective. The object view captures the destination but not the journey, and for grokking, the journey is the phenomenon.

Verdicts. *Process:* established—checkpoint analysis provides direct evidence of the phase transition. *Object:* valid but incomplete—the post-transition circuit is identified, but origin and timing remain unexplained.

B.6 World Models

Do language models build internal world models? Li et al. (2023b) show that linear probes recover Othello board state from intermediate activations of a model trained only on move sequences. But “encodes state” and “has a world model” are different claims under different views.

Subspace view. Linear probes recover board state with high accuracy. This is subspace evidence: the information exists in a linearly accessible subspace. It does not establish that the model *uses* this information causally—a model could encode state as a byproduct of next-move prediction without consulting it as a coherent representation.

Object view. Activation patching can identify specific components that mediate board-state information between layers. This establishes causal relevance of specific parts but not whether they compose into anything deserving the name “world model.”

Structural view. A world model requires the model’s internal causal graph to mirror the world’s causal structure—not just encoding individual variables but encoding the relationships between them. Current evidence (individual variable probes) does not test for structural isomorphism. The claim “has a world model” requires showing that the model’s internal causal dependencies recapitulate the game’s causal dependencies.

Role view. Individual components might play “state-tracking roles.” But a world model requires *coordinated* roles—multiple components maintaining a coherent state representation across time steps, updating it according to game rules. Coordinated role assignment has not been tested.

Verdicts. *Subspace*: established—linear encodings robustly found. *Object*: partially established—some causal evidence via patching. *Structural*: not established—causal graph isomorphism not demonstrated. *Role*: not established—coordinated role assignment not tested.

B.7 Circuit Disagreements

ACDC (Conmy et al., 2023), EAP (Syed et al., 2023), and activation patching (Wang et al., 2023) find different circuits for the same task. On IOI alone, published circuits disagree on which attention heads are included, which edges matter, and how large the circuit is. This looks like a replication crisis but is actually a view-level category error.

Object view. If circuits are objects—real subgraphs of the computational graph—then at most one can be correct. Different methods use different intervention types, thresholds, and search procedures. Under the object view, this is genuinely problematic: there is no principled way to adjudicate without an independent criterion for circuit identity.

Role view. Each method identifies components that play the relevant functional role *under its own intervention semantics*. ACDC finds components whose removal degrades performance; EAP finds components with high gradient-weighted activation flow; activation patching finds components that mediate between clean and corrupted inputs. These are different roles, and there is no reason they should select the same components. Disagreement reveals implicit identity differences, not replication failure.

Perspectival view. Each method is a different measurement perspective. ACDC, EAP, and patching define “part of the circuit” through different formal operations. Disagreement is expected and informative—it tells you about the methods’ measurement assumptions as much as about the model. Cross-method convergence, where it occurs, is the strongest evidence precisely because it survives perspective changes.

Verdicts. *Object*: problematic—at most one circuit can be correct, but none has epistemic priority. *Role*: diagnostic—disagreement reveals implicit differences in identity criteria. *Perspectival*: expected—method-dependence is the prediction, not the failure mode.

B.8 Probe Features: The Probing Wars Dissolved

Does a model “use” a linguistic feature just because a probe can decode it? The “probing wars” of 2019–2022 (Hewitt & Manning, 2019; Belinkov, 2022) debated whether probes discover real representations or impose structure. The framework suggests it was a view disagreement disguised as a methodological one.

Instrumental view. “Uses” means a probe can predict the feature above chance. This is the weakest notion—decodability, not causal use. A sufficiently powerful probe can decode almost any feature from high-dimensional activations.

Perspectival view. “Uses” means the feature is not a measurement artifact. Requires discriminant validity: the probe must find structure absent in control conditions (random models, shuffled activations).

Subspace view. “Uses” means the feature occupies a causal subspace—intervening on it changes behavior. DAS and IIA provide this evidence: they identify subspaces such that swapping activations transfers feature-specific behavior between inputs.

Role view. “Uses” means the feature plays a functional role—the model *needs* it. This is the strongest notion: removing or disrupting the feature degrades task performance in a way that cannot be compensated. Functional necessity is rarely tested.

Verdicts. *Instrumental*: established—probes work, which was never disputed. *Perspectival*: partially addressed—some control comparisons conducted. *Subspace*: emerging—DAS provides initial causal evidence. *Role*: rarely tested—functional necessity remains the gap. The probing wars were not a methodological disagreement about probe complexity. They were a view disagreement: “decodable” (instrumental) vs. “causally used” (subspace) vs. “functionally necessary” (role).

B.9 Decomposition Identity

Different decomposition methods—SAEs at varying widths, NMF, PCA, ICA—extract different features from identical activations. Which decomposition is “real”?

Object view. There is a fact of the matter: the model has true features and the correct decomposition finds them. The difficulty is circularity—the “correct” decomposition is defined as the one recovering the “true” features, which are defined as whatever the “correct” decomposition finds. Without an independent criterion for feature identity, the Object view cannot adjudicate.

Subspace view. The subspace is real; the basis is gauge freedom. Different decompositions producing different basis vectors for similar subspaces are choosing different coordinates for the same geometric object. The test: compute subspaces spanned by each method’s top- k features and measure pairwise Grassmannian distances. This converts an unanswerable metaphysical question into a measurable geometric one.

Perspectival view. Each decomposition is a measurement perspective. No perspective is privileged. The productive questions are where they agree (convergent validity) and where they disagree (what each perspective reveals that others miss). Decomposition dependence is predicted, not anomalous.

Verdicts. *Object*: underdetermined—no principled selection criterion. *Subspace*: the resolution—subspace convergence across methods is measurable. *Perspectival*: the meta-lesson—decomposition dependence is expected, and convergent validity across methods is the standard.

B.10 Cross-Model Identity and Circuit Universality

“The same feature in different models” is among the most consequential claims in mechanistic interpretability, underpinning the universality hypothesis: that different networks converge on shared computational primitives. Three recent empirical studies make the view-dependence of this claim concrete. Merullo et al. (2024) find 78% head overlap between the IOI and Colored Objects circuits in GPT-2 Medium. Tigges et al. (2024) show that individual head identities are unstable across training (heads swap in and out) while the algorithm remains stable across both training and scale in the Pythia suite. Wang et al. (2024) find an average cross-architecture feature similarity (MPPC) of 0.74 between Pythia-160M (Transformer) and Mamba-130M (SSM) using SAE features, and identify an induction circuit in Mamba that is structurally analogous to the Transformer version—but with a characteristic “off-by-one” positional shift caused by Mamba’s local convolution layer.

Object view. “Same” means same component—same head index, same layer position. Merullo et al. (2024) provide the strongest object-level evidence: 25 of 32 top-2% heads are shared between IOI and Colored Objects in GPT-2 Medium. But the result is fragile across scale: in GPT-2 Large, only 5 of 10 top heads overlap; in GPT-2 XL, essentially none do. Tigges et al. (2024) confirm this directly—in Pythia-160M, name-mover head (4,6) acquires its behavior at 4×10^9 tokens and loses it at 3×10^{10} tokens. Object-level identity is model-size-dependent and training-time-dependent. Across architectures, it is logically impossible: there is no natural correspondence between head 7.3 in a 12-layer Transformer and any component in a 24-layer SSM.

Role view. “Same” means same functional role. This is where the empirical evidence is most robust. Merullo et al. (2024) show that both IOI and Colored Objects use the same three-step algorithm (duplication detection, inhibition/content gathering, mover copying), and the mover role is preserved even when specific heads change. Tigges et al. (2024) demonstrate that while individual heads swap in and out across training, the algorithm—S-inhibition, name-mover, copy-suppression—remains stable. In GPT-2 XL, where object overlap approaches zero, 4 of the top heads across both tasks are still functionally mover heads. Role transport holds even where object transport fails.

Subspace view. “Same” means same causal subspace, measurable via Grassmannian distance after alignment. Wang et al. (2024) provide indirect evidence: 30%+ of Pythia SAE features find a Mamba match with MPPC > 0.95, and matched features appear at proportionally equivalent depths (Pythia layer l maps to approximately Mamba layer $2l$). Merullo et al. (2024) speculate that mover heads operate in a shared subspace regardless of task (they write in the direction of attended token embeddings), but do not measure subspace similarity directly. Explicit Grassmannian distance measurements between cross-model feature subspaces have not been published.

Structural view. “Same” means same gauge orbit—the computation is identical up to reparameterization. This is where the evidence is most revealing. The IOI and Colored Objects circuits share node-level overlap (object) and algorithmic structure (role), but the *edge structure differs*: IOI uses inhibition (negative signal) while Colored Objects uses content gathering (positive signal) to drive the same mover heads. The topology is similar but not identical (Merullo et al., 2024). The Mamba induction circuit is structurally analogous to the Transformer version but has an off-by-one positional shift (Wang et al., 2024)—same algorithm, different implementation mechanics. Structural transport in the strict sense (gauge-orbit identity) fails in every case examined, even when role transport succeeds.

Verdicts. *Object*: fragile—78% overlap within a single model (Merullo et al., 2024), near-zero across scales (Tigges et al., 2024), impossible across architectures. *Role*: robust—the strongest level at which universality holds; functional roles persist across tasks, training, scale, and architectures. *Subspace*: promising but untested—high feature similarity across architectures (Wang et al., 2024) but no direct Grassmannian measurements. *Structural*: fails—edge structure and implementation mechanics differ even when the algorithm is “the same.” The universality hypothesis is well-supported at the Role level and unsupported at the Structural level. Most published universality claims implicitly assume Structural transport on the basis of Role evidence—a coherence violation the framework diagnoses precisely.

B.11 Chain-of-Thought Faithfulness

Does chain-of-thought reasoning reflect a model’s actual computation? Safety-critical: if CoT is unfaithful, monitoring reasoning traces for deceptive planning is unreliable (Lanham et al., 2023). Different views impose different standards on “faithful.”

Instrumental view. “Faithful” means CoT predicts behavior: adding or modifying steps changes the answer. Turpin et al. (2023) show biased few-shot examples produce biased CoT, which produces biased answers. But instrumental faithfulness is weak: a post-hoc rationalization that correlates with the answer would also pass this test.

Role view. “Faithful” means CoT tokens causally mediate computation, not merely correlate with the answer. Requires showing specific tokens are necessary for specific downstream computations. Systematic role-level tests have not been conducted.

Structural view. “Faithful” means the CoT computation graph is isomorphic to internal processing. The strongest notion—the one safety monitoring implicitly requires. No existing work attempts this comparison.

Perspectival view. CoT faithfulness looks different from different measurement perspectives. Cross-method convergence—do probing, patching, and CoT monitoring agree?—is the critical test, and it has not been demonstrated.

Verdicts. *Instrumental*: partially demonstrated—manipulations change outputs. *Role*: untested—no systematic causal tests. *Structural*: not attempted—no computation graph comparison. *Perspectival*: the critical gap—no cross-method convergence demonstrated.

B.12 Deceptive Alignment

Could a model pursue misaligned goals while appearing aligned during evaluation (Hubinger et al., 2019)? Detecting strategically deceptive behavior requires evidence far above what current methods provide.

Instrumental view. Behavioral tests—evaluation suites, adversarial prompting—are instrumental evidence. But competent deception is by definition behaviorally identical to genuine alignment on the evaluation distribution. Instrumental evidence is structurally insufficient: it catches incompetent deception but not the threat model that motivates concern.

Object view. Mechanistic interpretability could identify a “deception circuit.” But this faces circularity: identifying the circuit requires labeled examples of deception, which requires already knowing the model is deceptive. Without independent access to ground-truth deception, circuit discovery cannot get started.

Structural view. A deceptive mechanism should be detectable as a gauge-invariant feature—identifiable regardless of parameterization. This is the minimum standard: if the mechanism is visible only under a specific basis or specific inputs, it is not robustly identified. Current methods do not reliably achieve gauge-invariant circuit identification even for benign circuits.

Process view. Deception is defined partly by origin: the model must have acquired the mechanism during training in a way reflecting goal-directed optimization for concealment. Process evidence—when did the mechanism form? does its emergence correlate with capability jumps?—is necessary because a structurally identical circuit arising from benign memorization is not deceptive.

Verdicts. *Instrumental*: structurally insufficient—deceptive behavior is identical to aligned behavior by construction. *Object*: circular—finding the circuit requires knowing deception exists. *Structural*: minimum standard—gauge-invariant identification is the floor. *Process*: necessary—training-time formation evidence is required.

B.13 Safety Evidence Gaps

Prominent safety findings—refusal direction ablation, representation engineering for truthfulness—provide instrumental evidence (a direction controls a behavior) used for structural conclusions (the direction *is* the mechanism). The gap between these claims is where safety monitoring is most vulnerable.

Instrumental view. Steering vectors that suppress refusal establish that a direction is a sufficient behavioral lever. But sufficiency without necessity means many directions may be sufficient levers for the same behavior, and a sufficient lever need not correspond to the mechanism the model uses during normal operation.

Object view. Identifying which components implement refusal requires patching at the component level. The minimum for safety monitoring: necessity (removing the component removes refusal across distributions) *and* sufficiency (the component alone produces refusal). Most findings establish at most one.

Subspace view. Is the refusal direction a stable causal subspace? The critical test: does the same subspace remain causally active on out-of-distribution prompts and adversarial perturbations? A direction validated only on the discovery distribution provides no safety guarantee against distribution shift.

Structural view. The strongest requirement: does the refusal mechanism survive reparameterization? If the direction is basis-dependent, an adversary could fine-tune the model to rotate it away—preserving the gauge-invariant computation while defeating direction-based monitoring.

The evidence ladder for safety is: (1) instrumental—a lever exists; (2) object—specific components are necessary and sufficient; (3) subspace—the mechanism is stable across distributions; (4) structural—the mechanism is gauge-invariant.

Verdicts. *Instrumental:* demonstrated—steering reliably controls behavior. *Object:* partial—some patching evidence, necessity rarely established. *Subspace:* untested—cross-distribution stability not measured. *Structural:* not attempted—gauge invariance of safety mechanisms unknown.

B.14 Validation Methodology

Three foundational threats undermine MI evidence: validation circularity (evaluating on the discovery distribution), interpretability illusions (distribution-dependent feature labels), and absence of random baselines (arbitrary directions receive plausible labels) (Casper et al., 2024). The framework reveals these as symptoms of a single underlying issue—conflation of method projections with model properties.

Perspectival view. All three threats are predicted by this view. If features are method projections, then: (1) they explain the discovery distribution by construction; (2) labels change with distribution because the projection selects different aspects; (3) random directions receive plausible labels because labeling imposes narrative structure on high-dimensional signals. The Perspectival view identifies these as structural consequences, not oversights.

Object view. All three threats are solvable: held-out evaluation addresses circularity, multi-distribution labeling addresses illusions, comparison to random directions addresses baselines. The solutions are straightforward once the threats are acknowledged.

Subspace view. Subspace stability across distributions—same Grassmannian point on discovery and held-out data—provides a natural validation criterion stronger than label agreement and weaker than full causal validation. The expected Grassmannian distance between a learned subspace and a random subspace is known analytically, giving a null distribution for significance testing.

Verdicts. *Perspectival:* diagnostic—predicts all three threats as structural consequences. *Object:* prescriptive—standard validation practices directly address the diagnosed threats. *Subspace:* measurable—cross-distribution Grassmannian stability is a formal, testable criterion with known null distributions.

C Full Axis Assignments

Table 4 repeats the five-axis assignments from Table 1 and adds the characteristic failure mode for each view.

View	Ontology	Identity	Evidence	Formalism	Target	Failure mode
Object	Concrete part	Component overlap	Ablation, patching	Directed graph	Specific behavior	Role inflation; distribution sensitivity
Role	Functional role	Role equivalence	Role-specific causal tests	Functional decomp.	Functional class	Role proliferation; underdefined roles
Subspace	Causal subspace	Same projector	DAS/IIA (linear)	$\text{Gr}(k, d)$	Representational variable	Unconstrained-alignment vacuousness; SAE conflation
Structural	Gauge-invariant structure	Gauge orbit	Holonomy, composition	Fiber bundle	Computation class	False gauge invariance; arch. prior confound
Process	Formation trajectory	Trajectory type	Checkpoints, knockouts	Dynamical system	Mechanism origin	Process-circuit conflation; training overfitting
Stratified	Stratum point	Stratum + local equiv.	Dimensionality, rank	Whitney strat.	Resolution-relative	Stratum misidentification from method artifacts
Perspectival	Method projection	Cross-method coherence	Multi-method robustness	Measurement algebra	Method-relative	Anti-realist slide; perspectival conflation
Instrumental	Predictive model	Predictive equiv.	Forecast, intervention	Model theory	Behavior prediction	Safety-relevant overreach
Contrastive	Difference relative to foil	Same contrastive pattern	Foil-varied patching	Contrastive explanation	Why P rather than Q	Foil dependence; contrastive cherry-picking

Table 4: Full axis assignments and characteristic failure modes.

D Open Problems in Mechanistic Interpretability

We surveyed seven sources spanning hundreds of individual open questions in mechanistic interpretability (Sharkey et al., 2025; Nanda, 2022; Balesni et al., 2024; Schmidt Futures, 2026; Steinhardt, 2026; Williams et al., 2025; Mueller et al., 2025) and distilled 17 distinct problem types. Each maps onto specific view-level confusions, ceilings, or under-explored territories. Of the 17: 8 are **clarified** (the question becomes answerable once the view is specified), 2 are **reframed** (the framework clarifies what evidence would answer it), and 7 are **scoped** (the framework identifies which view owns the problem).

D.1 Clarified Problems

These questions become answerable once the view is specified.

Decomposition identity. SAE features change with dictionary width—which decomposition is “real”? Under the object view, this is a problem: the enumeration depends on resolution. Under the subspace view, it is expected: the *subspace* spanned by a cluster of split features is what is real; how you decompose it is gauge freedom. SAEs finding features in noise (Bricken et al., 2023) is a perspectival-view prediction: decompositions are partly method artifact, requiring cross-method validation.

SAE “true” features. “Do SAEs recover the true features?” presupposes an object-view ontology in which discrete features are the fundamental units. Under the subspace view, the question becomes whether the subspace converges across widths and seeds—measurable via Grassmannian distance.

Superposition. “How do models represent more features than dimensions?” is a problem under the object view (feature entanglement to decompose), a geometric fact under the subspace view (expected overlap), and a presupposition under the perspectival view (it assumes discrete features exist to be “superposed”).

Circuit disagreements. ACDC, EAP, and activation patching find different circuits for the same task. Under the object view, at most one can be correct. Under the role view, each method identifies components

that play the relevant role under its own intervention semantics—they need not agree, and disagreement is diagnostic of different implicit identity criteria.

Unit of analysis. Are attention heads, neurons, or subspaces the right unit? The unit depends on the view: object view uses architectural components, role view uses functional roles, subspace view uses causal subspaces.

Probe features. “Does the model use feature X ?” means four different things: decodable (instrumental), more than artifact (perspectival), causally active (subspace), or functionally necessary (role).

Cross-model identity. “Same features across models” requires specifying the identity criterion. Same top-activating examples (instrumental) is weak. Same ablation effect (object) is architecture-dependent. Same subspace (subspace) is measurable via Grassmannian distance. Same gauge orbit (structural) is the strongest criterion. Most universality claims use instrumental evidence for structural conclusions.

DAS in random models. Unconstrained DAS achieving high IIA on random models (Sutter et al., 2025) is a perspectival-view prediction: unrestricted nonlinear alignment always finds structure, so what it finds in a real model needs stronger validation.

D.2 Reframed Problems

These questions remain empirically open, but the framework clarifies what evidence would answer them.

CoT faithfulness. “Is chain-of-thought faithful?” is four questions: does CoT predict behavior (instrumental), converge with other methods (perspectival), describe real functional roles (role), or map onto the computational graph (structural)? The safety-relevant notion requires cross-method convergence that no one has demonstrated.

World models. Current evidence that models encode state predicates linearly is subspace-level (the encoding exists). Whether these encodings form a causal world model—in which the model’s internal causal graph mirrors the world’s—is a structural claim requiring evidence of causal relationships between the encoded variables, not just their individual existence.

D.3 Scoped Problems

These are real and structural, arising from the limits of the view being used.

Intervention limits. Ablation-based methods face three Object-view ceilings: reconfiguration (the remaining network compensates), interaction effects (single-component ablation misses synergies), and side effects (ablating multi-functional components conflates necessity with general stability). Moving to the subspace view (orthogonal interventions) or structural view (gauge-invariant properties) partially resolves each.

Linear assumption. Circular and manifold representations challenge the assumption that concepts are encoded as linear directions. This is a subspace-view ceiling: the current formalism assumes linear subspaces, but the evidence increasingly shows nonlinear structure.

Safety evidence gaps. Prominent safety findings (refusal direction ablation, representation engineering for truthfulness) are instrumental evidence: they establish that a direction is a sufficient behavioral lever, not that it is the mechanism. Safety monitoring requires at minimum object evidence (necessity plus sufficiency) and ideally subspace evidence (stability across distributions).

Validation methodology. Three foundational threats to MI evidence: validation circularity (testing on the discovery distribution), interpretability illusions (distribution-dependent feature labels), and lack of

random baselines (arbitrary directions receive plausible labels). All are predicted by the perspectival view and resolved by requiring convergent evidence across methods and distributions.

Training dynamics. Most MI analyzes static snapshots, but mechanisms form and can be destroyed during training. Fine-tuning silently destroys circuits; adversarial training can relocate mechanisms; phase transitions create mechanisms suddenly. These belong to the process view, which uniquely provides evidence for cross-seed reliability and robustness to continued training.

Resolution and completeness. “How much do we understand?” is ill-posed without specifying resolution. Known circuits explain nearly everything at coarse granularity but effectively nothing at fine granularity. This is stratified-view territory: the interesting objects are the stable strata and the structure that survives coarse-graining.

Deceptive alignment. Detecting deceptive alignment requires evidence far above what current methods produce. Instrumental evidence (steering vectors, probes) is structurally insufficient: competent deception is behaviorally identical to alignment. The minimum standard is structural evidence (gauge-invariant deception circuit) combined with process evidence (formation trajectory during training).

D.4 Resolution Summary

Problem	Status	Key views
Decomposition identity	Clarified	Object vs. Subspace
SAE “true” features	Clarified	Object vs. Subspace
Superposition	Clarified	Object vs. Subspace vs. Perspectival
Circuit disagreements	Clarified	Object vs. Role
Unit of analysis	Clarified	Object vs. Role vs. Subspace
Probe features	Clarified	Instrumental vs. Subspace vs. Role
Cross-model identity	Clarified	Object impossible, Subspace possible
DAS in random models	Clarified	Perspectival diagnosis
CoT faithfulness	Reframed	Instrumental vs. Structural
World models	Reframed	Subspace vs. Structural
Intervention limits	Scoped	Object ceiling
Linear assumption	Scoped	Subspace ceiling
Safety evidence gaps	Scoped	Instrumental floor
Validation methodology	Scoped	Perspectival diagnosis
Training dynamics	Scoped	Process territory
Resolution/completeness	Scoped	Stratified territory
Deceptive alignment	Scoped	Structural + Process required

Table 5: Resolution status of 17 open problems across 7 sources. Clarified: the question becomes answerable once the view is specified. Reframed: the framework clarifies what evidence would answer it. Scoped: the framework identifies which view owns the problem.